

# Modeling infectious diseases using R

## Practical Session:

### Static and Dynamic aspects of SIR model

Ziv Shkedy, Rashider Aloni, Leyla Kodalci  
Center for Statistics, Hasselt University, Belgium

First Training workshop, Cape Town, South Africa  
12/10/2026-16/12/2026



Sustainable Training in Epidemiological Analysis and Modelling  
for Data-Driven Decision Making in Africa

# What do we cover in this practical session ?

- Simple transmission models in R.
- SIR: static aspect (time homogeneity).
- SIR: dynamic aspects.
- Vaccination.
- Software:
  - R.
  - The **deSolve** package in R.

# R requirements

- Basic knowledge in R.
- Programming of a user defined simple function.
- Why R ?
  1. Free.
  2. Fast (not in our case).
  3. Updated.
  4. Documented.

# R programs for the practical session

- HTML file for slides 5-78:
  - [eR\\_prog1\\_STEAM3D\\_2026\\_V1.html](#)
- Two R programs for slides 5-78:
  - R program:
    - [ModelingIDinR1\\_V1\\_Stat&Dynam\\_10\\_2026.R.](#)
    - R requirements: basic R editor.
  - R markdown:
    - [eR\\_prog1\\_STEAM3D\\_2026\\_V1.Rmd.](#)
    - R requirements: R Studio.

Part 1:  
Time homogeneity (the static model)

# Interactive tutorial: the HTML file

The screenshot shows a web browser window with the address bar displaying the file path: `C:/Ziv_Temp_2023/TanyLee_STEAM-3D/Workshop1_12_10_26/ShortCourse/Practicalessions_New/er_prog1_STEAM3D_2026-V1.html`. The browser interface includes a search bar, a "Find" button, and a "Publish" button. On the left side, there is a sidebar with a table of contents:

- 1. The SIR model
- 2. The SIR model in a closed population
- 3. The SIR model in open population
- 4. Threshold and Equilibrium Values

The main content area displays the date "10-09-2026" and the author "Ziv Shkedy and Rashider Aloni based on Chapter 2-4 in the book *Modeling infectious diseases parameters based on serological and social contact data* (<https://www.openintro.org/book/biostat/>)". The title of the tutorial is "Modeling infectious diseases using R - the SIR model". Below the title, there is a "Show" button. The first section is "1. The SIR model", which begins with a paragraph: "The starting point in this chapter is the SIR model in endemic equilibrium. In particular, we assume that all parameters are constant over time and the focus goes to the **age** of the individual as the time scale of primary interest. In Section XXX equilibrium values for the SIR-model with vital dynamics and a constant population were derived in case of homogeneous mixing. In practice however age is a predominant factor in the way an infection spreads through a population given that children have had fewer years of exposure than adults."

The next section is "Formulation of the SIR model", which starts with the text: "Extending the SIR model in terms of changes over **age** and **time** yields the following system of partial differential equations:".

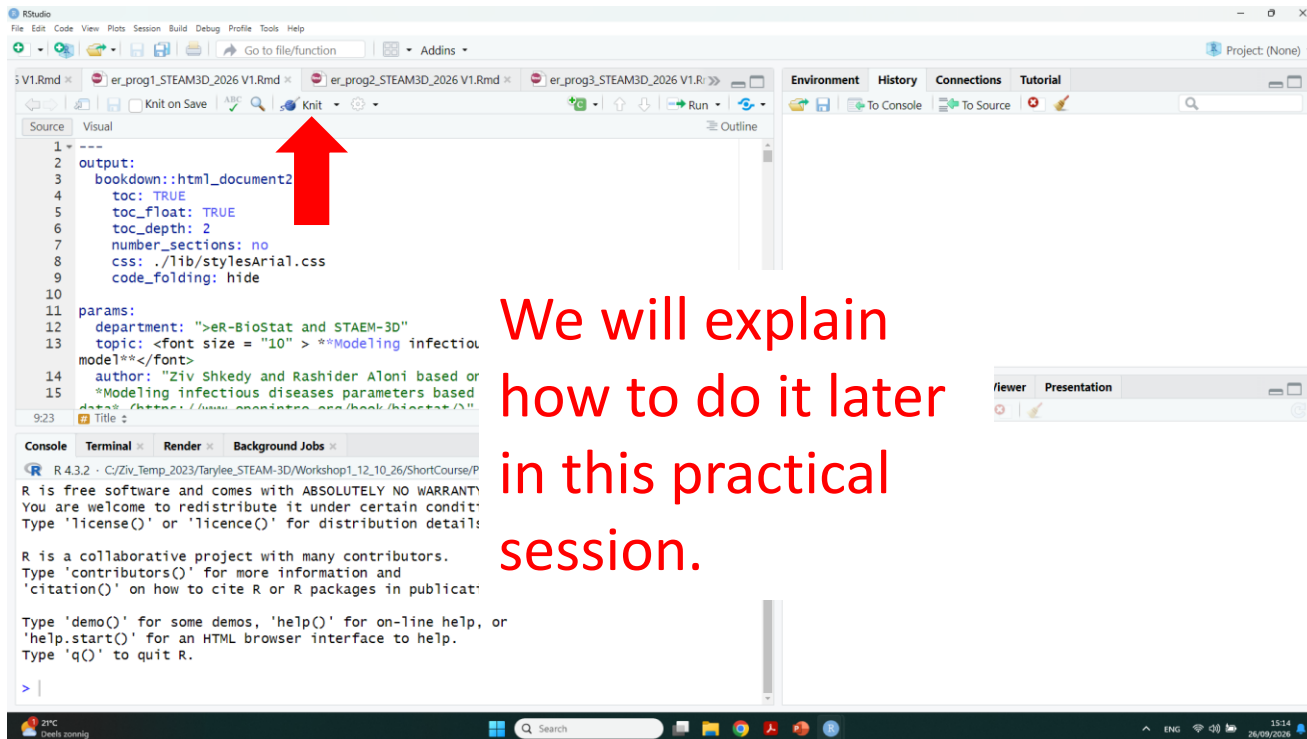
At the bottom of the page, the mathematical equation for the partial differential equation is partially visible:

$$\frac{\partial S(a, t)}{\partial t} + \frac{\partial S(a, t)}{\partial a} = -\mu(a)S(a, t) + \lambda(a)S(a, t)$$

Section 1 & 2 in the HTML file.

# How to produce the HTML file ?

- Open R Studio.
- Open the file: eR\_prog1\_STEAM3D\_2026\_V1.Rmd.
- Knit the file.



# SIR model :time homogeneity

$$\frac{dS(a)}{da} = -\lambda S(a)$$

$$\frac{dI(a)}{da} = \lambda S(a) - \sigma I(a)$$

$$\frac{dR(a)}{da} = \sigma I(a)$$



The force of infection: the rate in which individuals are infected



The recovery rate: the rate in which individuals recovered and move to the immune class.

Assumption: life long immunity.

# SIR model: time homogeneity

The diagram shows three differential equations for the SIR model. A red arrow points from the parameter  $\lambda$  in the first equation to its interpretation. Another red arrow points from the parameter  $\sigma$  in the third equation to its interpretation.

$$\frac{dS(a)}{da} = -\lambda S(a)$$

The force of infection = 0.2.  
On average: 5 years in the susceptible class.

$$\frac{dI(a)}{da} = \lambda S(a) - \sigma I(a)$$
$$\frac{dR(a)}{da} = \sigma I(a)$$

The recovery rate: 10 days.

The unit of the parameters are in years

# Transmission models in R

- We need to integrate the system of the ordinary differential equation.
- deSolve package in R.
- Numerical integration using of ODE system.

$$\begin{aligned}\frac{dS(a)}{da} &= -\lambda S(a) \\ \frac{dI(a)}{da} &= \lambda S(a) - \sigma I(a) \\ \frac{dR(a)}{da} &= \sigma I(a)\end{aligned}$$

# Transmission models in R

## SIR model

$$\begin{aligned}\frac{dS(a)}{da} &= -\lambda S(a) \\ \frac{dI(a)}{da} &= \lambda S(a) - \sigma I(a) \\ \frac{dR(a)}{da} &= \sigma I(a)\end{aligned}$$

## Specification in R

- Model parameters.
- State variables (the value of the parameters at age (time) zero.
- Time range (=age range) for integration.

# Transmission models in R

## R program

Model parameters



$\lambda, \sigma$

State variables



$S(0), I(0), R(0)$

Time range for  
integration



Integrate from age zero  
to age 40

Specification of the  
model



$$\begin{aligned}\frac{dS(a)}{da} &= -\lambda S(a) \\ \frac{dI(a)}{da} &= \lambda S(a) - \sigma I(a) \\ \frac{dR(a)}{da} &= \sigma I(a)\end{aligned}$$

# Specification of the model parameters in R

```
> parameters <- c(lambda = 0.2, v=36.5)
> parameters
lambda      v
0.2      36.5
```

The force of  
infection = 0.2  
years.

$$5^{-1} = 0.2$$

Recovery rate 10  
days.

$$\left(\frac{10}{365}\right)^{-1} = 36.5$$

The unit of the  
parameters are  
in years<sup>-1</sup>

# The state variables (initial values at age 0)

- Let us assume that the cohort size is 5000.
- At age=0:

$$S(0) = 4999$$

$$I(0) = 1$$

$$R(0) = 0$$

- Specification in R:

```
> state <-  
c(X=4999, Y=1, Z=0)  
> state  
      X      Y      Z  
4999    1      0
```

# Specification of the model in R

$$\left. \begin{aligned} \frac{dS(a)}{da} &= -\lambda S(a) \\ \frac{dI(a)}{da} &= \lambda S(a) - \sigma I(a) \\ \frac{dR(a)}{da} &= \sigma I(a) \end{aligned} \right\}$$

```
SIR<-  
function(t,state,parameters)  
{  
  with(as.list(c(state,  
parameters))),  
  {  
    dX <- -lambda*X  
    dY <- lambda*X - v*Y  
    dZ <- v*Y  
    list(c(dX, dY, dZ))  
  })  
}
```

We ask from the function to  
return the values of S, I and R <sup>15</sup>

# Specification of the time units for the integration

$$\begin{aligned}\frac{dS(a)}{da} &= -\lambda S(a) \\ \frac{dI(a)}{da} &= \lambda S(a) - \sigma I(a) \\ \frac{dR(a)}{da} &= \sigma I(a)\end{aligned}$$

- The solution of the model: numerical integration.
- Time units: age.
- Integration from age 0 to age 40 by unit of 0.01 years

```
> times<-seq(0,40,by=0.01)
> times
[1] 0.00 0.01 0.02 0.03 0.04 0.05 0.06 0.07 0.08 0.09 0.10 0.11
[13] 0.12 0.13 0.14 0.15 0.16 0.17 0.18 0.19 0.20 0.21 0.22 0.23
[25] 0.24 0.25 0.26 0.27 0.28 0.29 0.30 0.31 0.32 0.33 0.34 0.35
[37] 0.36 0.37
```

# Running the model

```
require(deSolve)  
out<-as.data.frame(ode(y=state,times=times,func=SIR,parms=parameters))
```



The state variables:  
the values at age 0.

The model  
parameters: force of  
infection (0.2) and  
recovery rate (10  
days)

# Running the model

```
require(deSolve)  
out<-as.data.frame(ode(y=state,times=times,func=SIR,parms=parameters))
```



Time range for  
integration.



The model



Numerical integration  
using ordinary  
differential equation

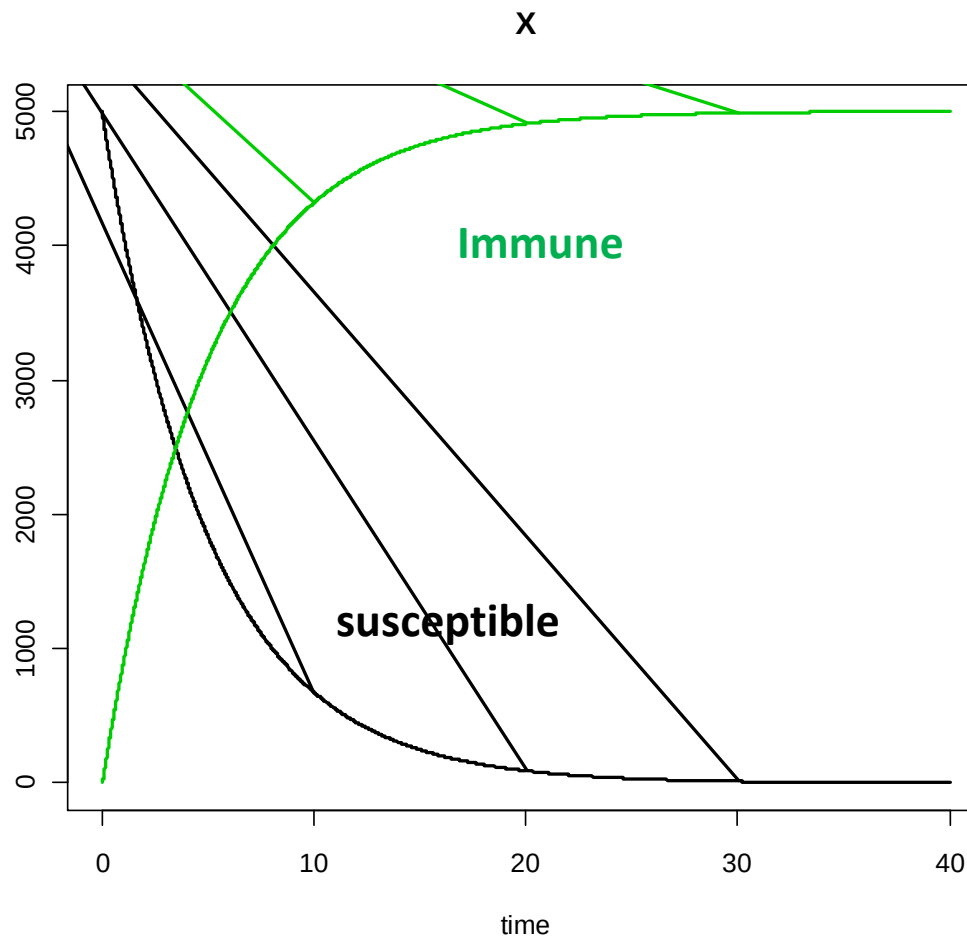
# Solution

```
> require(deSolve)
> out <-
as.data.frame(ode(y=state, times=times, func=SIR, parms=
parameters))
> head(out)
```

|   | time | X        | Y         | Z         |
|---|------|----------|-----------|-----------|
| 1 | 0.00 | 4999.000 | 1.000000  | 0.000000  |
| 2 | 0.01 | 4989.012 | 9.061818  | 1.926190  |
| 3 | 0.02 | 4979.044 | 14.641580 | 6.314481  |
| 4 | 0.03 | 4969.096 | 18.498345 | 12.405853 |
| 5 | 0.04 | 4959.168 | 21.159066 | 19.673391 |
| 6 | 0.05 | 4949.259 | 22.989501 | 27.751380 |

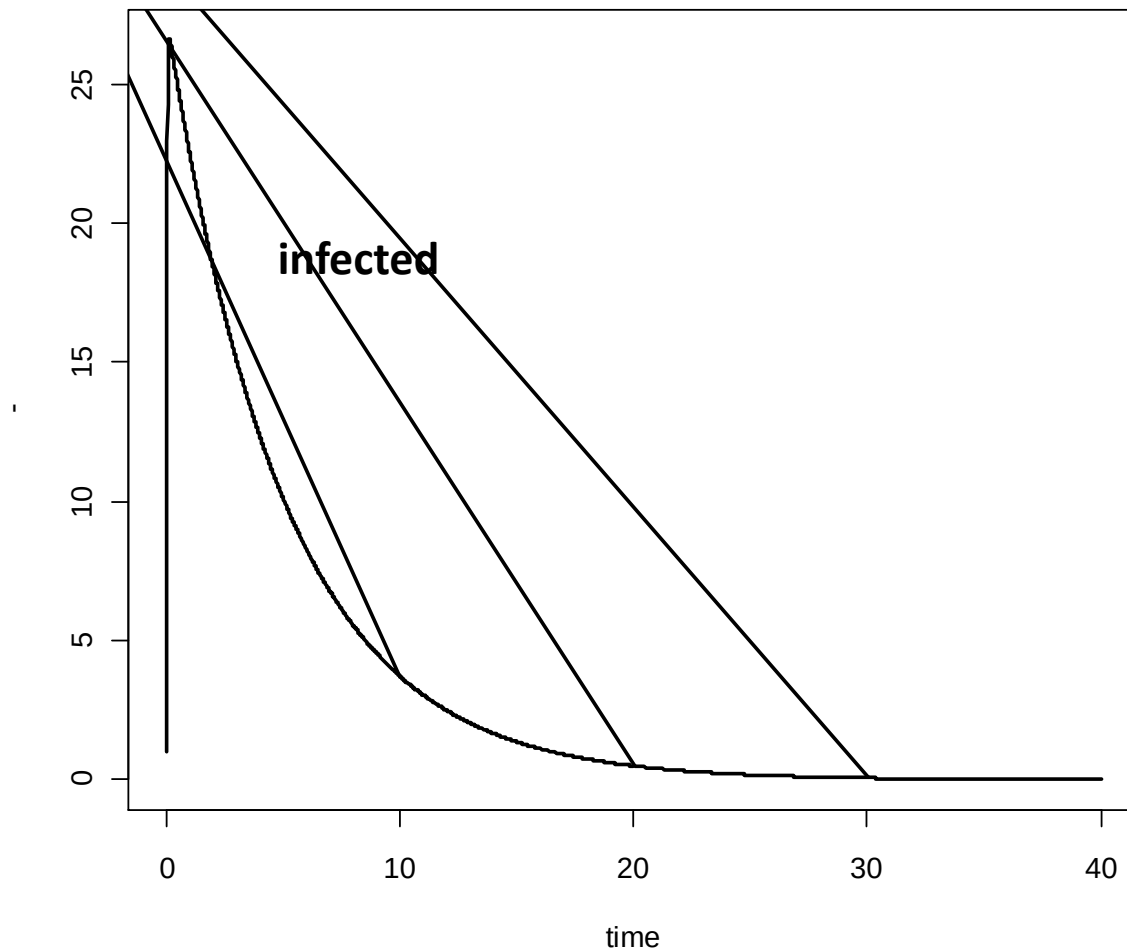
# Graphical output

```
> plot (times,out$X ,type="l",main="X", xlab="time", ylab="-",lwd=2)  
> lines(times,out$Z,col=3,lwd=2)
```



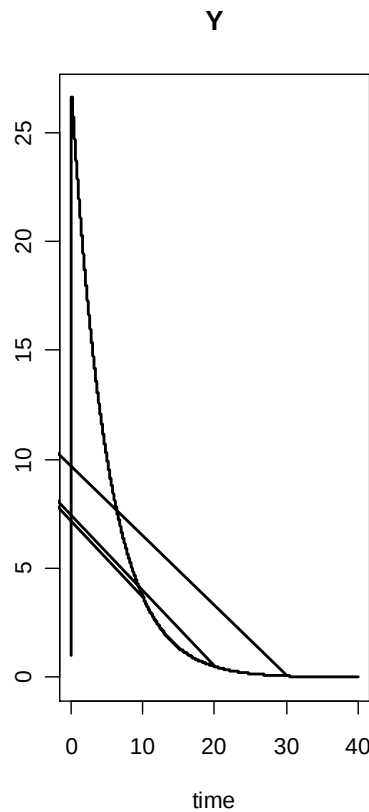
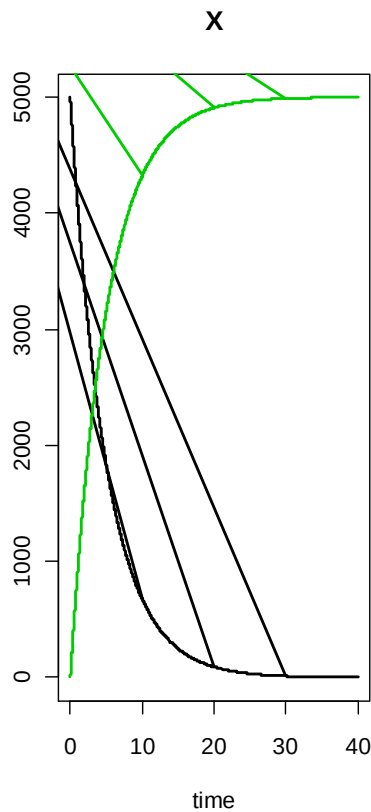
# Graphical output

```
> plot (times,out$Y ,type="l",main="Y", xlab="time", ylab="-",lwd=2)
```



# Graphical output

SIR model



What do we see here ?

Number of infected individuals at each age.

# Slides 13-23: example 1 in the HTML

C:/Ziv\_Temp\_2023/TanyLee\_STEAM-3D/Workshop1\_12\_10\_26/ShortCourse/Practicallessons\_New/er\_prog1\_STEAM3D\_2026-V1.html

er\_prog1\_STEAM3D\_2026-V1.html Open in Browser Find Publish

1. The SIR model

2. The SIR model in a closed population

Example 1 ( $\lambda = 0.2, D=10$  days)

Example 2 ( $\lambda = 0.2, D=60$  days)

Example 3: age dependent force of infection

3. The SIR model in open population

4. Threshold and Equilibrium Values

## Example 1 ( $\lambda = 0.2, D=10$ days)

SIR model with  $\lambda = 0.2$  and  $D = 10$  days ( $\nu = 36.5$ )

For a population in which the average age at infection is 5 years (i.e., the average time that is spent in the susceptible class is 5 years) we have

$$\lambda = \frac{1}{5} = 0.2.$$

For a recovery rate of 10 days,

$$\nu = \left(\frac{10}{365}\right)^{-1} = 36.5$$

In R, the object `parameters` is used to store the parameter values.

```
## lambda      v
##      0.2    36.5
```

We assume a population size of  $N = 5000$  with one infected individual at age=0,  $I(0) = 1$ . This implies that  $S(0) = N - 1 = 4999$  and  $R(0) = 0$ . The `state` vector is given by

```
##      X      Y      Z
## 4999      1      0
```

Note that the R objects `X`, `Y` and `Z` correspond to the compartments `S`, `I` and `R`, respectively.

# The HTML file

1. The SIR model

2. The SIR model in a closed population

Example 1 ( $\lambda = 0.2, D=10$  days)

Example 2 ( $\lambda = 0.2, D=60$  days)

Example 3: age dependent force of infection

3. The SIR model in open population

4. Threshold and Equilibrium Values

$$\frac{dR(a)}{da} = \nu I(a) - \mu(a)R(a),$$

## 2. The SIR model in a closed population

We consider a closed population, i.e., individuals do not enter or exit to/from the population, that is  $\mu(a) = 0$ . In this case the SIR model is given by

$$\begin{aligned}\frac{dS(a)}{da} &= -\lambda(a)S(a), \\ \frac{dI(a)}{da} &= \lambda(a)S(a) - \nu I(a), \\ \frac{dR(a)}{da} &= \nu I(a).\end{aligned}$$

The model

There are two parameters in the model, the force of infection  $\lambda$  and the recovery rate  $\nu$ .

### Model assumptions

- No mortality access due to the disease.
- Life long immunity.

### Example 1 ( $\lambda = 0.2, D=10$ days)

Show

### SIR model with $\lambda = 0.2$ and $D = 10$ days ( $\nu = 36.5$ )

For a population in which the average age at infection is 5 years (i.e., the average time that is spent in the susceptible class is 5 years) we have

$$\lambda = \frac{1}{D} = 0.2$$

# The HTML file: example 1

C:\Ziv\_Temp\_2023\Tanyee\_STEAM-3D\Workshop1\_12\_10\_26\ShortCourse\Practicalessions\_New\er\_prog1\_STEAM3D\_2026-V1.html

er\_prog1\_STEAM3D\_2026-V1.html Open in Browser Find Publish

- Life long immunity.

1. The SIR model

2. The SIR model in a closed population

**Example 1 ( $\lambda = 0.2, D=10$  days)**

Example 2 ( $\lambda = 0.2, D=60$  days)

Example 3: age dependent force of infection

3. The SIR model in open population

4. Threshold and Equilibrium Values

## Example 1 ( $\lambda = 0.2, D=10$ days)

Show

### SIR model with $\lambda = 0.2$ and $D = 10$ days ( $\nu = 36.5$ )

For a population in which the average age at infection is 5 years (i.e., the average time that is spent in the susceptible class is 5 years) we have

$$\lambda = \frac{1}{5} = 0.2.$$

For a recovery rate of 10 days,

$$\nu = \left(\frac{10}{365}\right)^{-1} = 36.5$$

In R, the object `parameters` is used to store the parameter values.

```
parameters <- c(lambda = 0.2, v=36.5)
parameters
```

Code

```
## lambda      v
##      0.2    36.5
```

We assume a population size of  $N = 5000$  with one infected individual at age=0,  $I(0) = 1$ . This implies that  $S(0) = N - 1 = 4999$  and  $R(0) = 0$ . The `state` vector is given by

Show

21°C Deels zonnig

Search

ENG 15:27 26/09/2026

# The HTML file: the model in R

CrZiv\_Temp\_2023/Tanylee\_STEAM-3D/Workshop1\_12\_10\_26/ShortCourse/Practicalessions\_New/er\_prog1\_STEAM3D\_2026-V1.html

er\_prog1\_STEAM3D\_2026-V1.html Open in Browser Find Publish

1. The SIR model

2. The SIR model in a closed population

Example 1 ( $\lambda = 0.2, D=10$  days)

Example 2 ( $\lambda = 0.2, D=60$  days)

Example 3: age dependent force of infection

3. The SIR model in open population

4. Threshold and Equilibrium Values

## Model formulation

The SIR model

$$\begin{aligned}\frac{dS(a)}{da} &= -\lambda(a)S(a), \\ \frac{dI(a)}{da} &= \lambda(a)S(a) - \nu I(a), \\ \frac{dR(a)}{da} &= \nu I(a),\end{aligned}$$

can be defined in R using the `SIR` function in the following way

```
SIR<-function(t,state,parameters)
{
  with(as.list(c(state, parameters)),
  {
    dX <- -lambda*X
    dY <- lambda*X - v*Y
    dZ <- v*Y
    list(c(dX, dY, dZ))
  })
}
```

Code for the model

The object `times` is the ages that we use to integrate the equation system in order to produce the output. In our example, the age ranges from 0 to 40 years old.

```
## [1] 0.00 0.01 0.02 0.03 0.04 0.05 0.06 0.07 0.08 0.09 0.10 0.11 0.12 0.13 0.14
## [16] 0.15 0.16 0.17 0.18 0.19
```

21°C  
Deels zonnig

Search

ENG 15:29  
26/09/2026

# The HTML file: running the model in R

1. The SIR model

2. The SIR model in a closed population

Example 1 ( $\lambda = 0.2, D=10$  days)

Example 2 ( $\lambda = 0.2, D=60$  days)

Example 3: age dependent force of infection

3. The SIR model in open population

4. Threshold and Equilibrium Values

Using the `ode()` function in the `deSolve` package for numerical integration.

## Running the model

We solve the equations system and produce the model's solution using the function `ode` of the R package `deSolve`. In the panel below,

$$S(0) = 4999, I(0) = 1, R(0) = 0,$$

and

$$S(0.05) = 4949.259, I(0.05) = 22.989501, R(0.05) = 27.751380.$$

## Numerical solution

The R object `out` contain the numerical solution of the model. The objects `X`, `Y` and `Z` are the number of susceptible, infected and immune individuals at age `a` (the R object `times`).

```
require(deSolve)
out <- as.data.frame(ode(y=state, times=times, func=SIR, parms=parameters))
head(out)
```

| ##   | time | X        | Y         | Z         |
|------|------|----------|-----------|-----------|
| ## 1 | 0.00 | 4999.000 | 1.000000  | 0.000000  |
| ## 2 | 0.01 | 4989.012 | 9.061818  | 1.926190  |
| ## 3 | 0.02 | 4979.044 | 14.641580 | 6.314481  |
| ## 4 | 0.03 | 4969.096 | 18.498345 | 12.405853 |
| ## 5 | 0.04 | 4959.168 | 21.159066 | 19.673391 |
| ## 6 | 0.05 | 4949.259 | 22.989501 | 27.751380 |

## Graphical solution

Figure 1 shows the numerical solution for the example. We notice that, the fraction infected individuals is relatively small compared to the fraction susceptible and immune. This is due to the different time spent in each compartment. The average duration in the susceptible class is 5 year and after recovery individuals gain life long immunity against reinfection.

# The HTML file

1. The SIR model

2. The SIR model in a closed population

Example 1 ( $\lambda = 0.2, D=10$  days)

**Example 2 ( $\lambda = 0.2, D=60$  days)**

Example 3: age dependent force of infection

3. The SIR model in open population

4. Threshold and Equilibrium Values

## Graphical solution

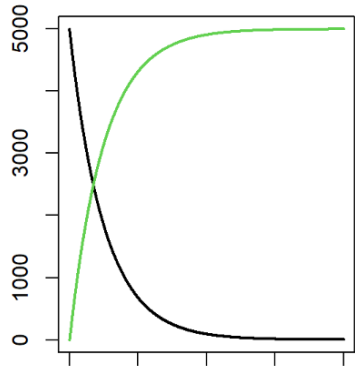
Figure 1 shows the numerical solution for the example. We notice that, the fraction infected individuals is relatively small compared to the fraction susceptible and immune. This is due to the different time spent in each compartment. The average duration in the susceptible class is 5 year and after recovery individuals gain life long immunity against reinfection.

```
par(mfrow=c(1,2), oma=c(0,0,3,0))
plot (times,out$X ,type="l",main="S and R", xlab="age", ylab="-",lwd=2)
lines (times,out$Z,col=3,lwd=2)
plot (times,out$Y ,type="l",main="I", xlab="time", ylab="-",lwd=2)
mtext(outer=TRUE,side=3,"SIR model, D=10 days",cex=1.5)
```

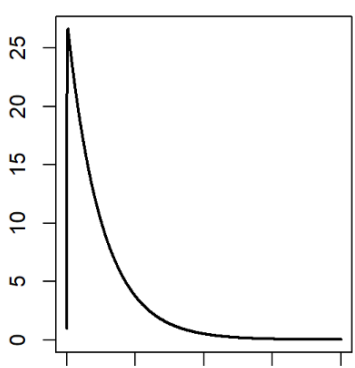
Code to produce the figures.

### SIR model, D=10 days

#### S and R



#### I

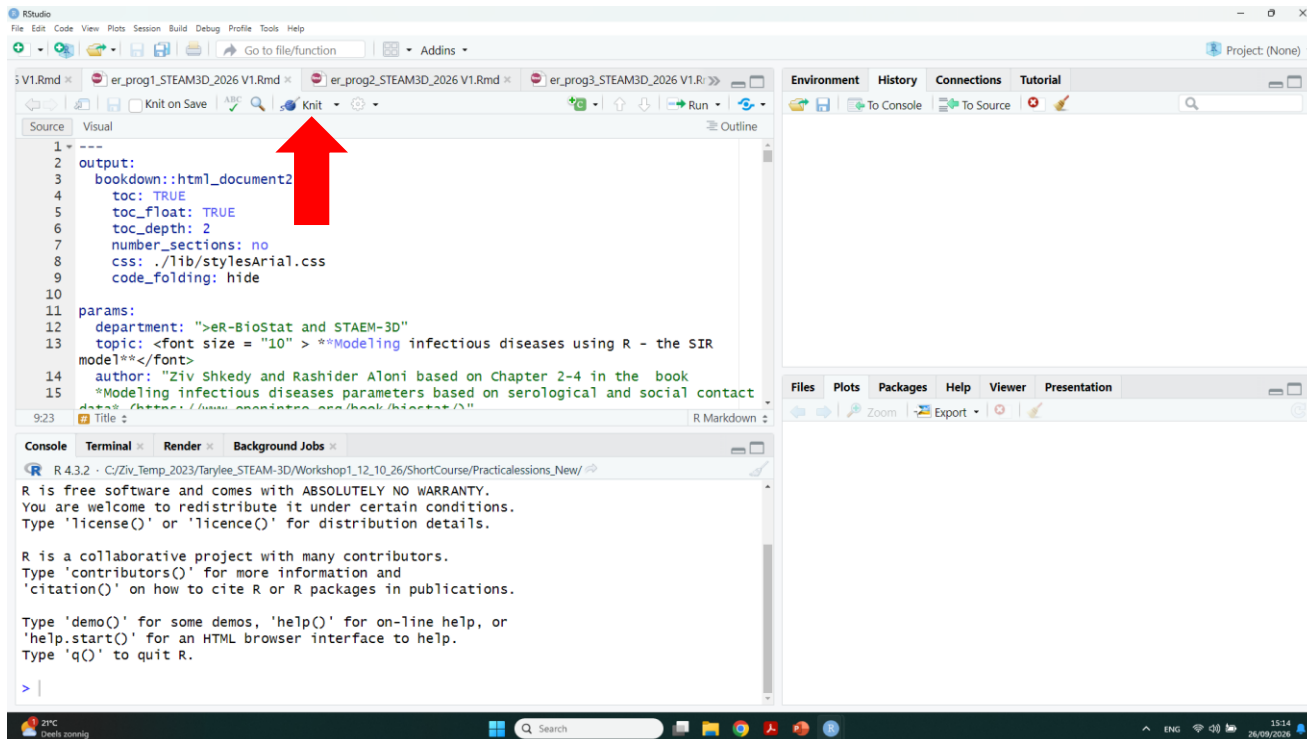


# Running the model in R

- Produce the model's output by yourself.
- Open R/R studio.
- Open the program:
  - [ModelingIDinR1\\_V1\\_Stat&Dynam\\_10\\_2026.R.](#)

# How to produce the HTML file ?

- Open R Studio.
- Open the file: `eR_prog1_STEAM3D_2026_V1.Rmd`.
- Knit the file.



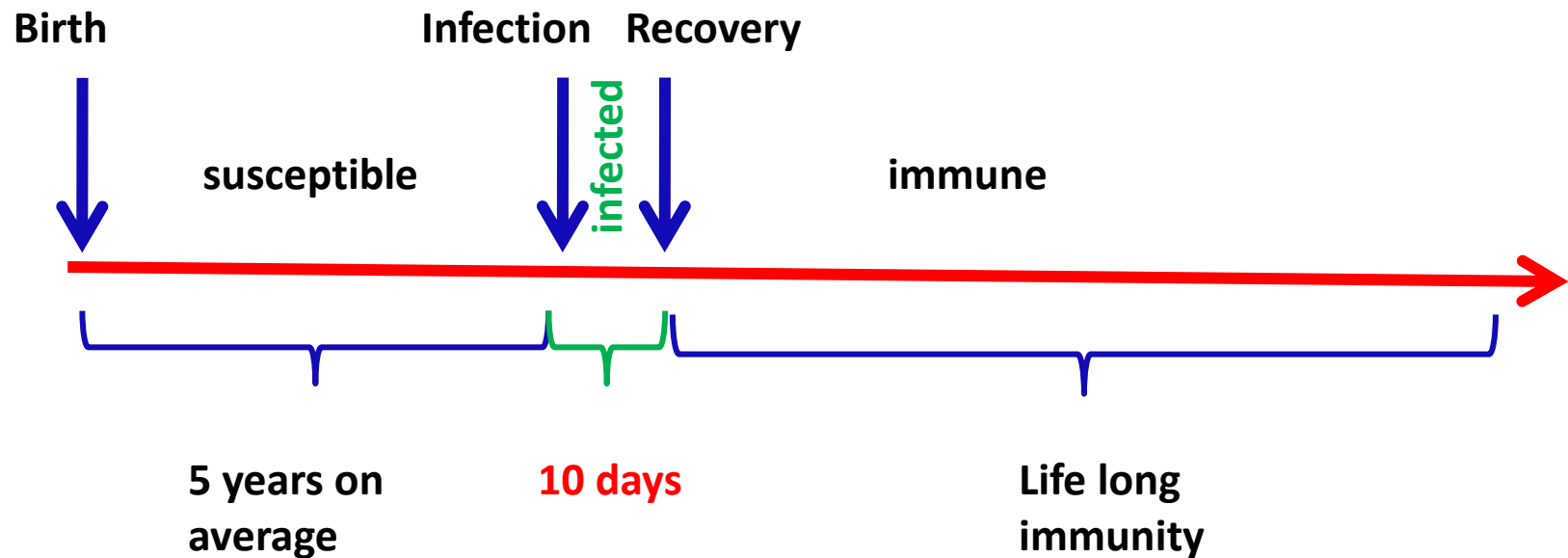
# Changing the parameter setting

- We run the model again but higher duration in the susceptible class.

$$D=10 \text{ days} \rightarrow D = 60 \text{ days}$$

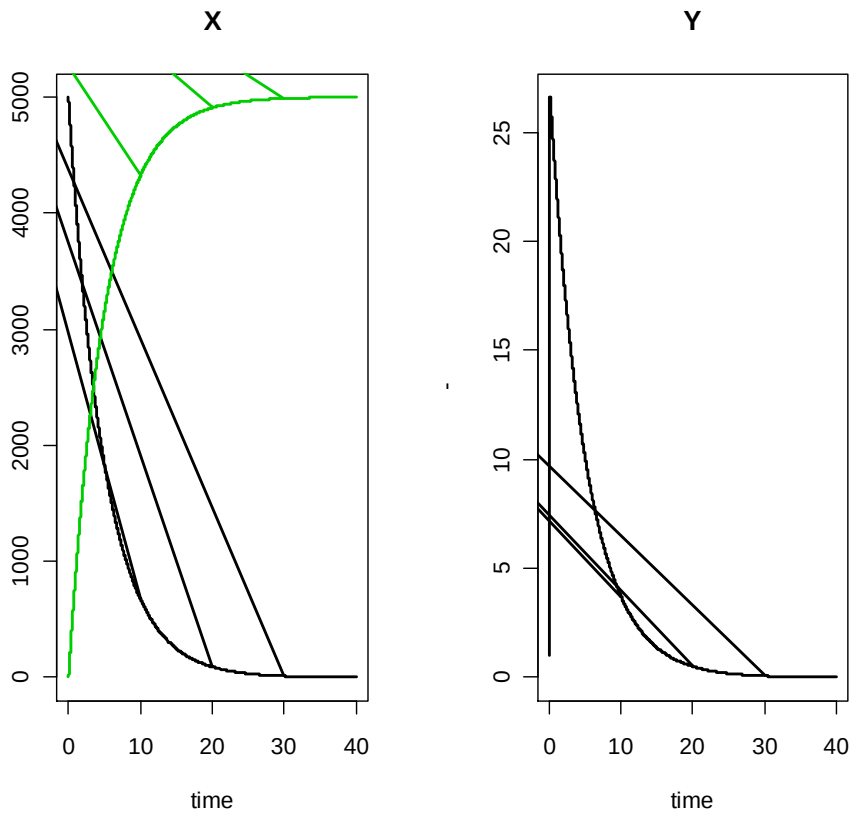
$$\left(\frac{10}{365}\right)^{-1} = 36.5 \quad \left(\frac{60}{365}\right)^{-1} = 6.08$$

# Duration of stay in the different compartments of the models



# Graphical output

SIR model



We expect to see only few infected individuals at each age (compared to the number of susceptible and immune).

# Slides 31-37: example 2 in the HTML file changing the recovery rate

1. The SIR model

2. The SIR model in a closed population

Example 1 ( $\lambda = 0.2, D=10$  days)

**Example 2 ( $\lambda = 0.2, D=60$  days)**

Example 3: age dependent force of infection

3. The SIR model in open population

4. Threshold and Equilibrium Values

Figure 1: Solution for a SIR model with  $\lambda=0.2$  and  $D=10$  days.

**Example 2 ( $\lambda = 0.2, D=60$  days)**

SIR with  $\lambda = 0.2$   $D=\text{two months}$  ( $\nu = 6.08$ )

In this example we change the recovery rate from 10 days to two months (60 days) so individuals stay in the infected class a longer time compare to the first example. In this case

$$\nu = \left(\frac{60}{365}\right)^{-1} = 6.08$$

Show

## [1] 0.1643836

Show

## [1] 6.083333

The new parameters vector is given by

Show

| lambda   | v        |
|----------|----------|
| 0.200000 | 6.083333 |

Show

Runing the model

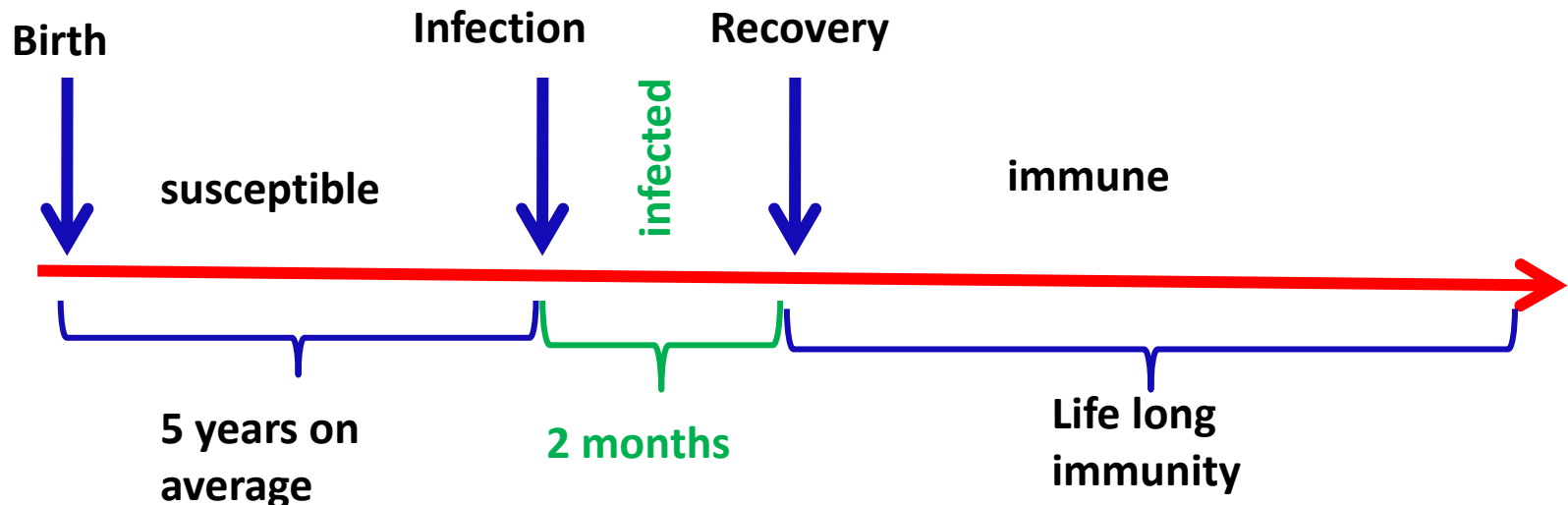
Note that compared to an average duration of 10 days in the infected class (Example 1), we expect to observed a high proportion of

$D=10 \text{ days} \rightarrow D = 60 \text{ day}$

# Duration of stay in the different compartments of the models

Let us assume that the recovery rate is 2 months (i.e. individuals stay in the infected class 2 months)

What do we expect to see ?



# The model with recovery rate of 2 months

1/(60/365)

```
parameters <- c(lambda = 0.2, v=6.083333)
```

```
parameters
```

```
state <- c(X=4999,Y=1,Z=0)
```

```
state
```

```
SIR<-function(t,state,parameters)
```

```
{
```

```
with(as.list(c(state, parameters)),
```

```
{
```

```
dX <- -lambda*X
```

```
dY <- lambda*X - v*Y
```

```
dZ <- v*Y
```

```
list(c(dX, dY, dZ))
```

```
})
```

```
}
```

```
times<-seq(0,40,by=0.01)
```

```
times
```

```
require(deSolve)
```

```
out <-
```

```
as.data.frame(ode(y=state,times=times,func=SIR,parms=parameters))
```

```
head(out)
```

```
par(mfrow=c(1,2), oma=c(0,0,3,0))
```

```
plot(times,out$X,type="l",main="X", xlab="time",  
ylab="-",lwd=2)
```

```
lines(times,out$Z,col=3,lwd=2)
```

```
plot(times,out$Y,type="l",main="Y", xlab="time",  
ylab="-",lwd=2)
```

```
mtext(outer=TRUE,side=3,"SIR model",cex=1.5)
```

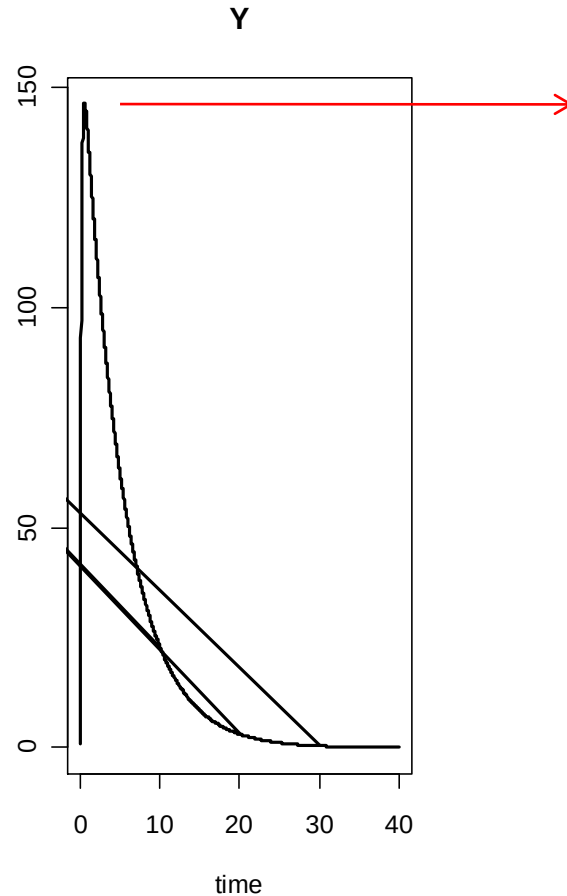
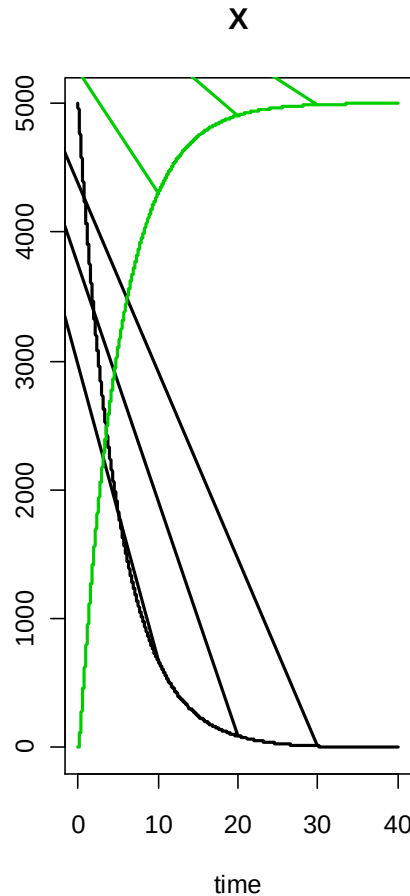

$$\left(\frac{60}{365}\right)^{-1} = 6.08$$

- Recover rate of 2 months
- (60 days).
- Code from the R program:
  - [ModelingIDinR1\\_V1\\_Stat&Dynam\\_10\\_2026.R](#)

# Graphical output

(force of infection of 0.2 and recovery rate of 2 months)

SIR model



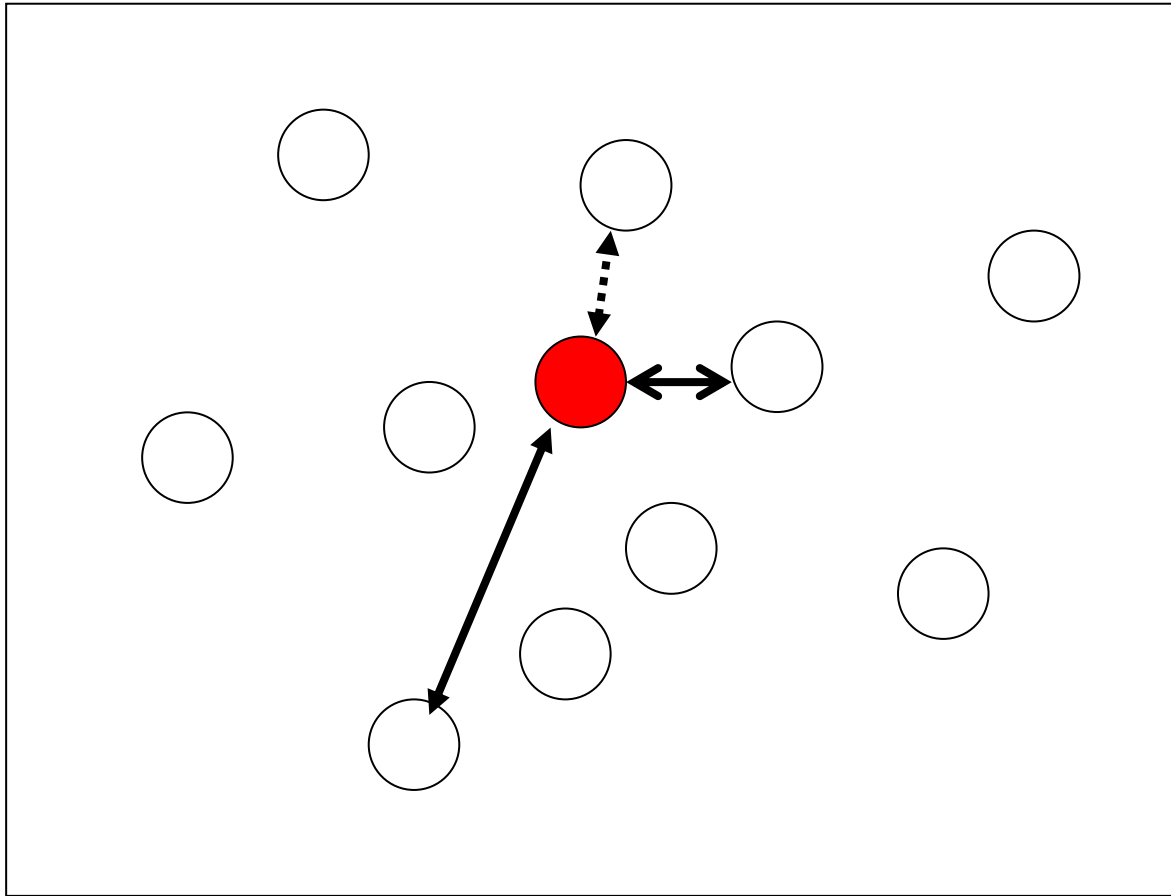
~ 150 infected individuals at the peak (compare with ~25 for recovery rate of 10 days)

# Running the model in R

- Produce the model's output by yourself.
- Open R/R studio.
- Open the program:
  - [ModelingIDinR1\\_V1\\_Stat&Dynam\\_10\\_2026.R.](#)

Part 2:  
The mass action principle (static model)

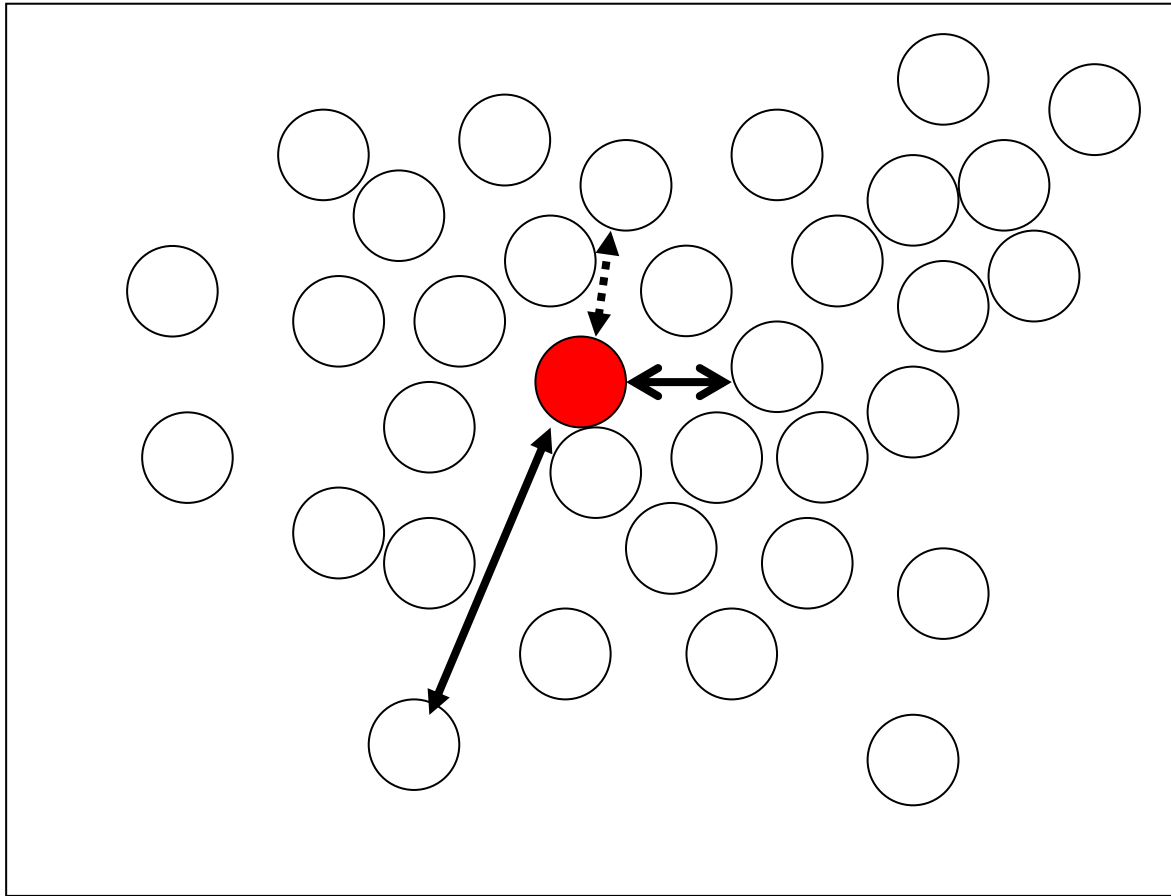
# The Mass-Action Principle



**Contacts are made in random.**

Number of new cases =  $P(\text{transmission}) \times \# \text{ of infectious} \times \# \text{ of susceptible}$

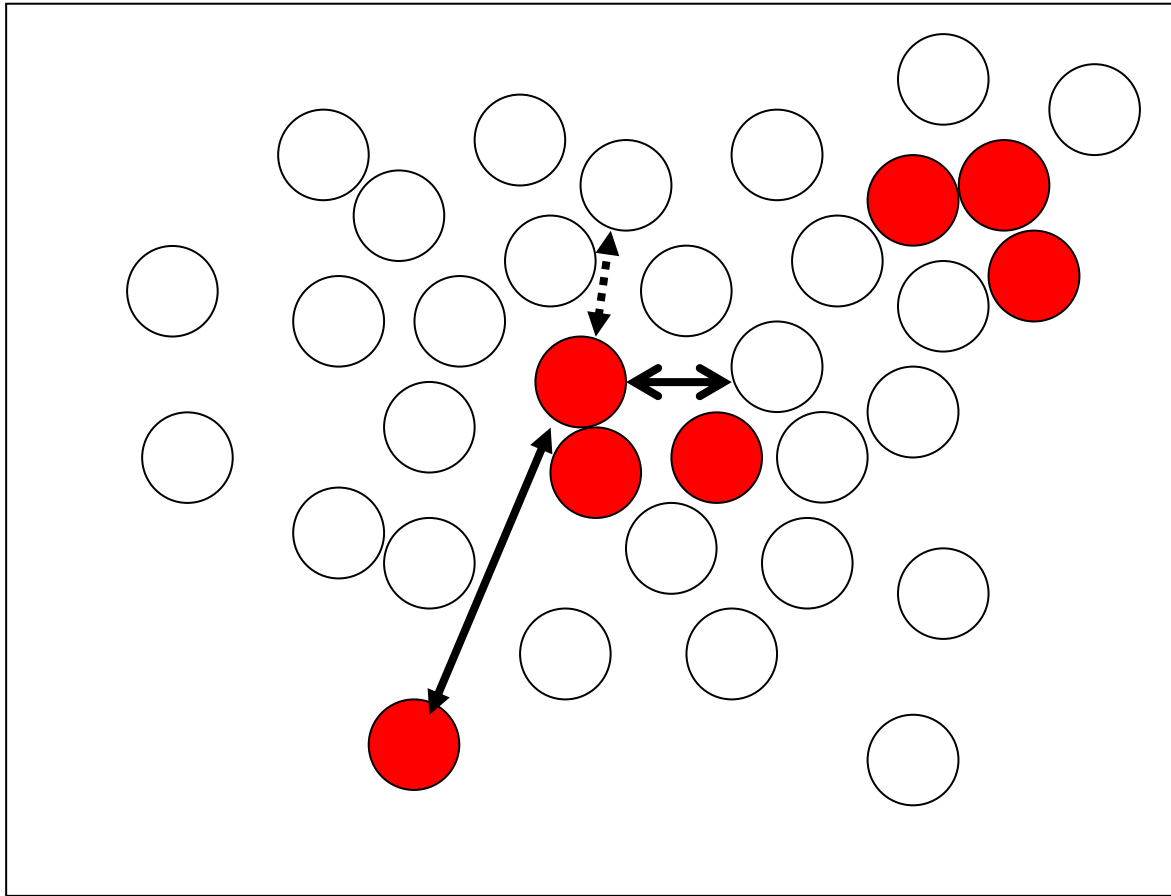
# The Mass-Action Principle



Contacts are made in random.

Number of new cases =  $P(\text{transmission}) \times \# \text{ of infectious} \times \# \text{ of susceptible}$

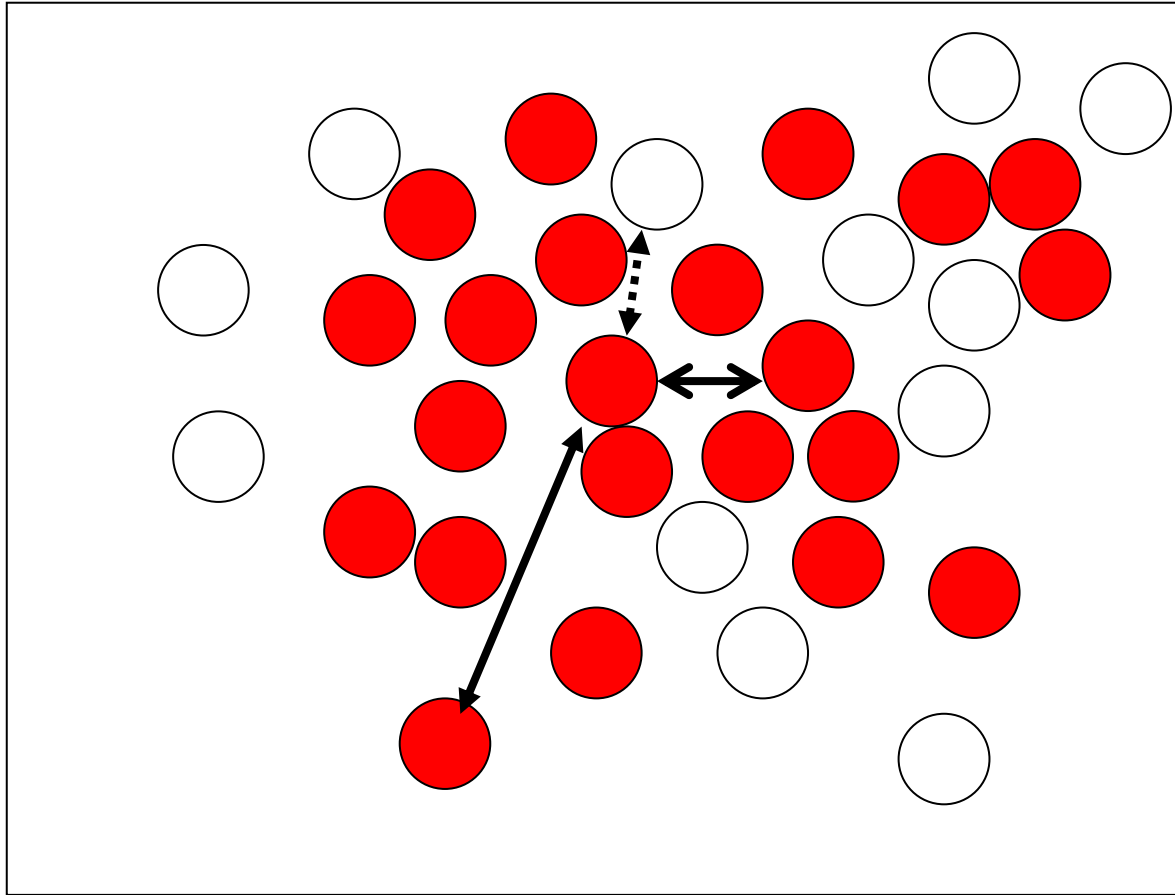
# The Mass-Action Principle



Contacts are made in random.

Number of new cases =  $P(\text{transmission}) \times \# \text{ of infectious} \times \# \text{ of susceptible}$

# The Mass-Action Principle



Contacts are made in random.

Number of new cases:

$$\beta \times I \times S$$

Transmission  
probability per  
contact

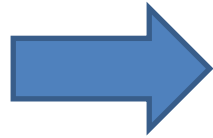
Number of new cases =  $P(\text{transmission}) \times \# \text{ of infectious} \times \# \text{ of susceptible}$

# The Mass-Action Principle

$$\frac{dS(a)}{da} = -\lambda S(a)$$

$$\frac{dI(a)}{da} = \lambda S(a) - \sigma I(a)$$

$$\frac{dR(a)}{da} = \sigma I(a)$$



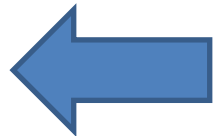
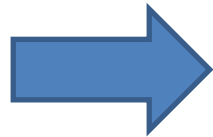
$$\frac{dS(a)}{da} = -\beta \times I(a) \times S(a)$$

$$\frac{dI(a)}{da} = \beta \times I(a) \times S(a) - \sigma I(a)$$

$$\frac{dR(a)}{da} = \sigma I(a)$$

# Age dependent force of infection

$$\begin{aligned}\frac{dS(a)}{da} &= -\lambda(a)S(a) \\ \frac{dI(a)}{da} &= \lambda(a)S(a) - \sigma I(a) \\ \frac{dR(a)}{da} &= \sigma I(a)\end{aligned}$$



$$\begin{aligned}\frac{dS(a)}{da} &= -\beta \times I(a) \times S(a) \\ \frac{dI(a)}{da} &= \beta \times I(a) \times S(a) - \sigma I(a) \\ \frac{dR(a)}{da} &= \sigma I(a)\end{aligned}$$

$$\lambda(a) = \beta \times I(a)$$

Age  
dependent

constant

# Model parameters

$$\begin{aligned}\frac{dS(a)}{da} &= -\beta \times I(a) \times S(a) \\ \frac{dI(a)}{da} &= \beta \times I(a) \times S(a) - \sigma I(a) \\ \frac{dR(a)}{da} &= \sigma I(a)\end{aligned}$$

Parameters in R:

```
> parameters <- c(beta=0.0085, v=36.5)
> state <- c(X=4999, Y=1, Z=0)
> parameters
  beta      v
0.0085 36.5000
> state
  X    Y    Z
4999  1    0
```

$$\begin{aligned}\beta &= 0.0085 \\ \sigma &= 36.5 \quad (10 \text{ days})\end{aligned}$$

10 days

$$\frac{10}{365} = 10 \text{ days in years}$$

$$\left(\frac{10}{365}\right)^{-1} = \left(\frac{365}{10}\right) = 36.5 = \sigma$$

Recovery rate = (duration in the infected class in years)<sup>-1</sup>

# Specification of the model in R

$$\begin{aligned}\frac{dS(a)}{da} &= -\beta \times I(a) \times S(a) \\ \frac{dI(a)}{da} &= \beta \times I(a) \times S(a) - \sigma I(a) \\ \frac{dR(a)}{da} &= \sigma I(a)\end{aligned}$$

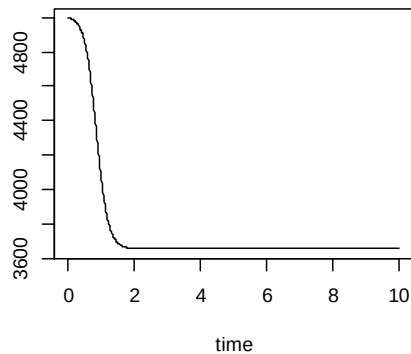


```
SIR<-function(t,state,parameters)
{
  with(as.list(c(state,
parameters))),
  {
    dX <- -beta*Y*X
    dY <- beta*Y*X - v*Y
    dZ <- v*Y
    list(c(dX, dY, dZ))
  })
}
```

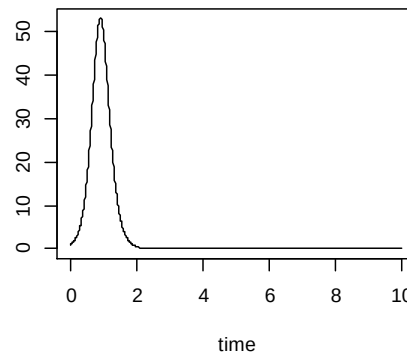
# Numerical solution

SIR model,  $\lambda = \beta \cdot I$

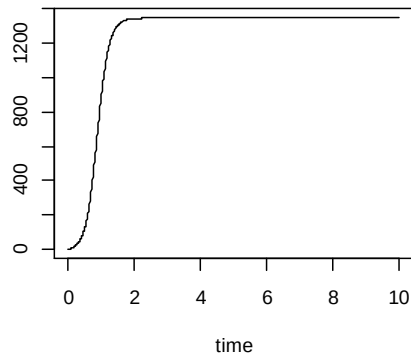
X



Y



Z



After  $\sim 2$  years there are no infected individuals.

$\sim 3600$  individuals will not be infected.

Equilibrium

# Slides 39-48: example 3 in the HTML file

C:/Ziv\_Temp\_2023/Tarjee\_STEAM-3D/Workshop1\_12\_10\_26/ShortCourse/Practicalsessions\_New/er\_prog1\_STEAM3D\_2026-V1.html

er\_prog1\_STEAM3D\_2026-V1.html Open in Browser Find Publish

Figure 2: Solutions for a SIR model D=60 days.

1. The SIR model  
2. The SIR model in a closed population  
Example 1 ( $\lambda = 0.2, D=10$  days)  
Example 2 ( $\lambda = 0.2, D=60$  days)  
**Example 3: age dependent force of infection**  
3. The SIR model in open population  
4. Threshold and Equilibrium Values

**Example 3: age dependent force of infection**

In this example we change the formulation of the force of infection and assume that  $\lambda(a) = \beta \times I(a)$ . Here  $\beta$  is the transmission rate and  $I(a)$  is the number of infected individuals at age  $a$ .

$$\beta = 0.0085 \text{ and } \lambda = \beta \times I(a)$$

For a recovery rate of 10 days, the new parameters vector is given by

Show

```
##      beta      v
## 0.0085 36.5000
```

Show

**Definition of the model**

For the SIR model, we use  $\beta \times Y \times X$  instead of  $\lambda \times X$

Show

**Runing the model and graphical output**

The solution of the model is presnted in Figure XXX. Note that the force of infection in this model is not constant but it is proportional for the number of individuals in the infected compartment presented in Figure 3.

Show

```
##      time      X      Y      Z
## 1 0.00 4999.000 1.000000 0.000000
## 2 0.01 4998.562 1.061727 0.3761535
## 3 0.02 4998.097 1.127220 0.7755187
```

21°C Grotendeels be... Search ENG 15:48 26/09/2026

# The HTML file: the model

C:/Ziv\_Temp/2023/Tarylee\_STEAM-3D/Workshop1\_12\_10\_26/ShortCourse/Practicallessons\_New/er\_prog1\_STEAM3D\_2026-V1.html

er\_prog1\_STEAM3D\_2026-V1.html Open in Browser

Find

Publish

## 1. The SIR model

### 2. The SIR model in a closed population

Example 1 ( $\lambda = 0.2, D=10$  days)

Example 2 ( $\lambda = 0.2, D=60$  days)

### Example 3: age dependent force of infection

## 3. The SIR model in open population

## 4. Threshold and Equilibrium Values

## Definition of the model

For the SIR model, we use  $\beta Y X$  instead of  $\lambda Y X$

```
SIR<-function(t,state,parameters)
{
  with(as.list(c(state, parameters)),
  {
    dX <- -beta*Y*X
    dY <- beta*Y*X - v*Y
    dZ <- v*Y
    list(c(dX, dY, dZ))
  })
}
```

Code for the model.

## Runing the model and graphical output

The solution of the model is presnted in Figure XXX. Note that the force of infection in this model is not constant but it is proportional for the number of individuals in the infected compartment presented in Figure 3.

```
##   time      X      Y      Z
## 1 0.00 4999.000 1.000000 0.0000000
## 2 0.01 4998.562 1.061727 0.3761535
## 3 0.02 4998.097 1.127220 0.7755187
## 4 0.03 4997.604 1.196705 1.1995104
## 5 0.04 4997.080 1.270419 1.6496279
## 6 0.05 4996.524 1.348611 2.1274607
```

# Running the model in R

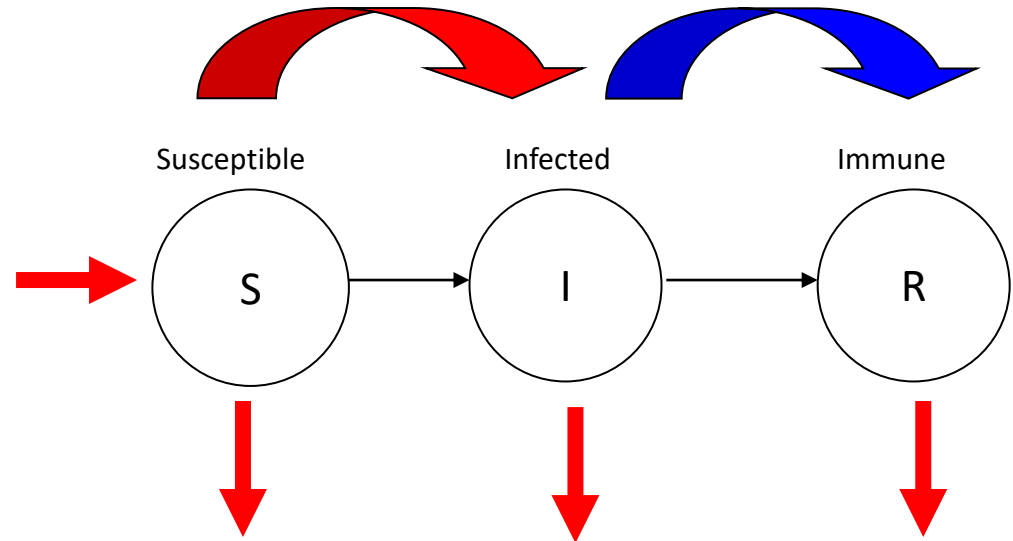
- Produce the model's output by yourself.
  - Open R/R studio.
  - Open the program:
    - [ModelingIDinR1\\_V1\\_Stat&Dynam\\_10\\_2026.R](#).

## Part 3:

# The SIR model in open population

# SIR transmission model in open population

$$\begin{aligned}\frac{dS(t)}{dt} &= N\mu - \lambda S - \mu S \\ \frac{dI(t)}{dt} &= \lambda S - (\sigma + \mu)I \\ \frac{dR(t)}{dt} &= \sigma I - \mu R\end{aligned}$$



Birth and death rate are equal to  $\mu$  (constant population size).

Population size:  $N$ .

Recovery rate:  $\sigma$ .

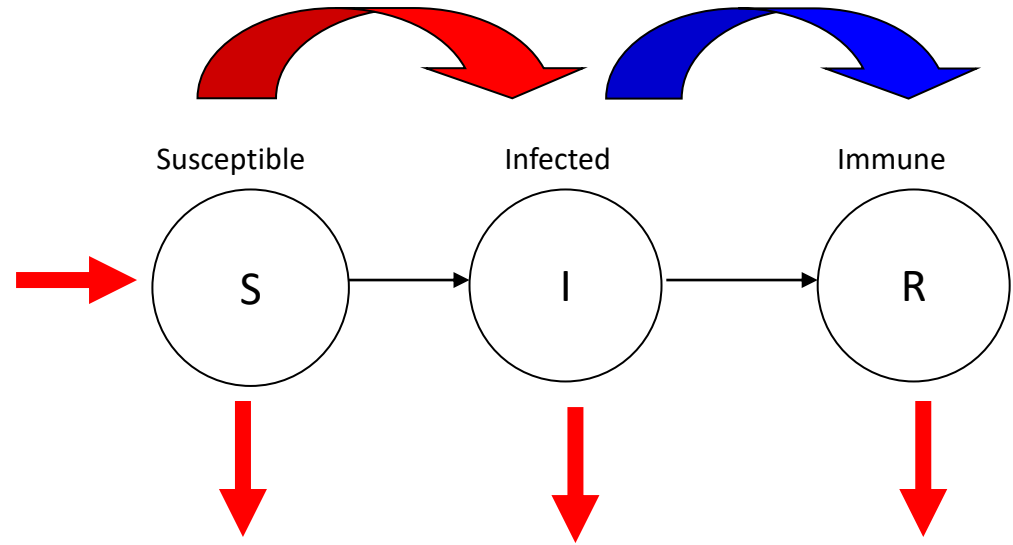
Force of infection:  $\lambda$ .

# SIR transmission model in open population

- The model in R

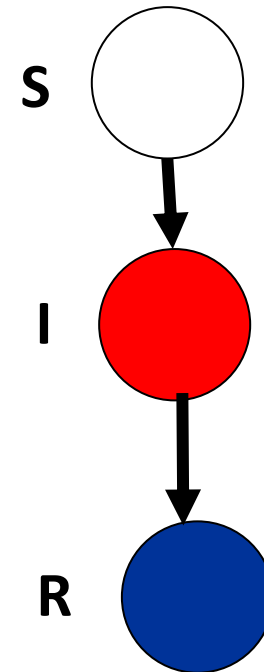
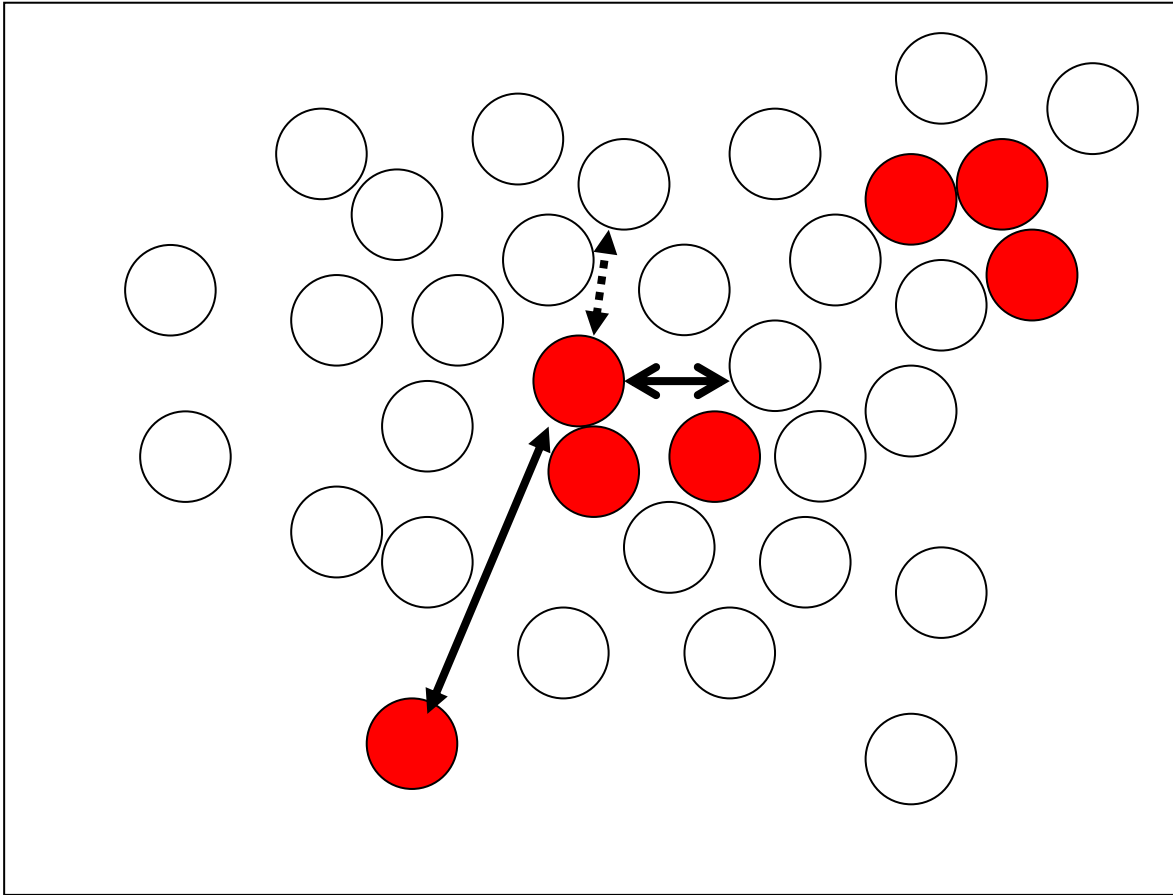
```
SIR<-function(t,state,parameters)
{
  with(as.list(c(state, parameters)),
  {
    dX <- 5000*mu-lambda*X-mu*X
    dY <- lambda*X - v*Y-mu*Y
    dZ <- v*Y-mu*Z
    list(c(dX, dY, dZ))
  })
}
```

- We assume that  $N=5000$ .



$$\begin{cases} \frac{dS(t)}{dt} = N\mu - \lambda S - \mu S \\ \frac{dI(t)}{dt} = \lambda S - (\sigma + \mu)I \\ \frac{dR(t)}{dt} = \sigma I - \mu R \end{cases}$$

# The law of mass action



New cases(t)=transmission rate X infected(t) X susceptible(t)

# SIR model in open population: Dynamic aspects

- In order to understand the dynamic of the SIR model we need to allow for time dependent force of infection.
- Open population.
- The force of infection (depends on time):

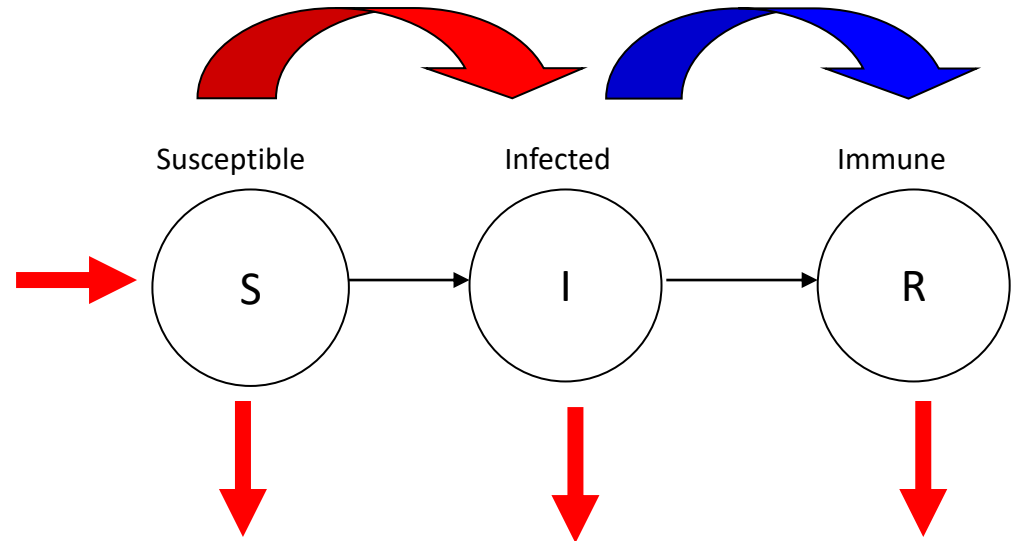
$$\lambda(t) = \beta \times I(t)$$

- Mass action principle.

$$\text{New case} = \beta \times I(t) \times S(t)$$

# SIR transmission model in open population

$$\begin{aligned}\frac{dS(t)}{dt} &= N\mu - \beta IS - \mu S \\ \frac{dI(t)}{dt} &= \beta IS - (\sigma + \mu)I \\ \frac{dR(t)}{dt} &= \sigma I - \mu R\end{aligned}$$



Birth and death rate are equal to  $\mu$  (constant population size).

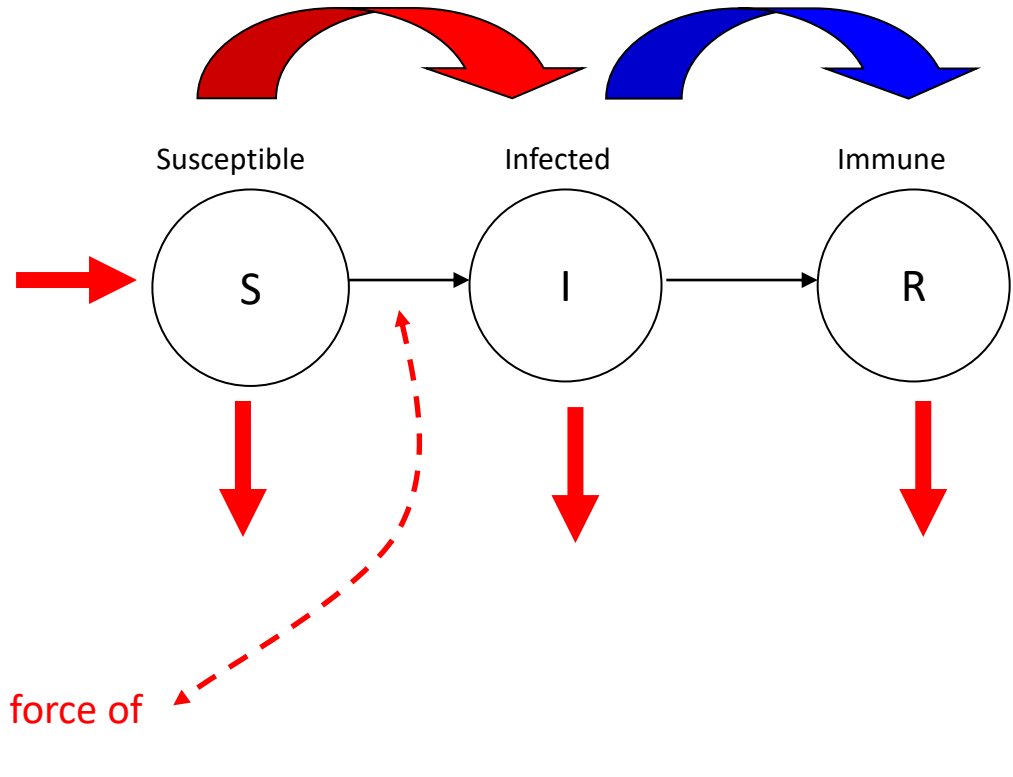
$N$ : population size.

$\sigma$ : recovery rate.

$\beta$ : transmission rate.

# SIR transmission model in open population

$$\begin{aligned}\frac{dS(t)}{dt} &= N\mu - \beta IS - \mu S \\ \frac{dI(t)}{dt} &= \beta IS - (\sigma + \mu)I \\ \frac{dR(t)}{dt} &= \sigma I - \mu R\end{aligned}$$

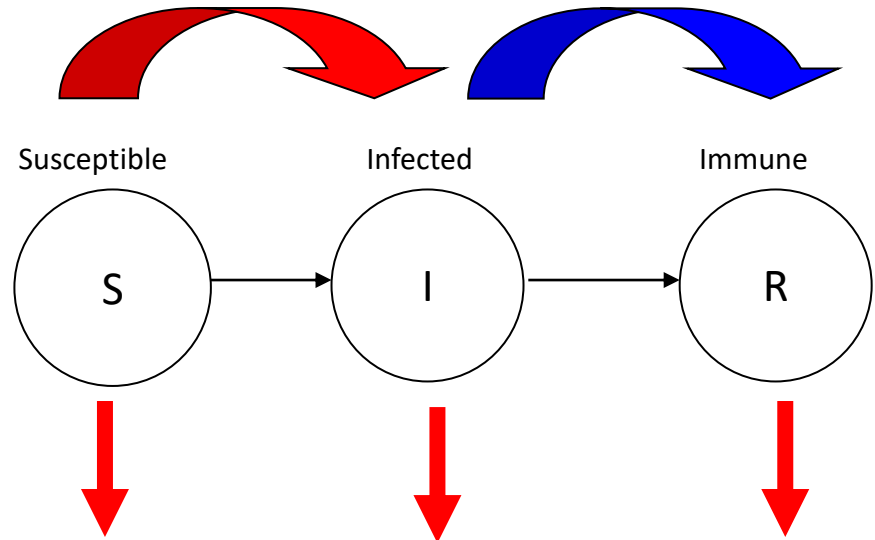


$\lambda(t) = \beta \times I(t)$ : force of infection  
 $\beta \times I(t) \times S(t)$ : number of new cases

# SIR transmission model in open population

- The model in R

```
SIR<-function(t,state,parameters)
{
  with(as.list(c(state, parameters)),
  {
    dX <- N*mu*(1-p)-beta*Y*X - mu*X
    dY <- beta*Y*X - v*Y - mu*Y
    dZ <- v*Y -mu*Z+N*mu*p
    list(c(dX, dY, dZ))
  })
}
```



- Note that  $p=0$  !!!

$$\left\{ \begin{array}{l} \frac{dS(t)}{dt} = N\mu - \beta IS - \mu S \\ \frac{dI(t)}{dt} = \beta IS - (\sigma + \mu)I \\ \frac{dR(t)}{dt} = \sigma I - \mu R \end{array} \right.$$

# Running the model in R

```
out <-  
as.data.frame(ode(y=state, times=times, func=SIR, parms=parameters))
```

- We need to provide:
  - The state variables.
  - Time for integration.
  - The parameter setting.
  - The function for the model.

# Parameters setting

- Parameter setting:

$\beta = \text{tranmission rate} = 0.001.$

$N = \text{population size} = 5000.$

$\mu = \text{birth rate} = \text{death rate} = 1/75.$

$v = \text{recovery rate} = 1.$

$p = \text{vaccinate rate} = 0.$

- Parameter setting in R:

```
> parameters <- c(mu=1/75, beta=0.001, v=1)
```

```
> print(parameters)
```

| mu         | beta       | v          |
|------------|------------|------------|
| 0.01333333 | 0.00100000 | 1.00000000 |

# State variables and time for integration

- The state vector:

$$(S(0) = N - 1 = 4999, I(0) = 1, R(0) = 0)$$

- The state vector in R:

```
> state <- c(X=4999, Y=1, Z=0)
> print(state)
      X      Y      Z
4999      1      0
```

# Population size and time for integration

- Population size:

```
> p<-0.0  
> p  
[1] 0  
> N<-5000  
> N  
[1] 5000
```

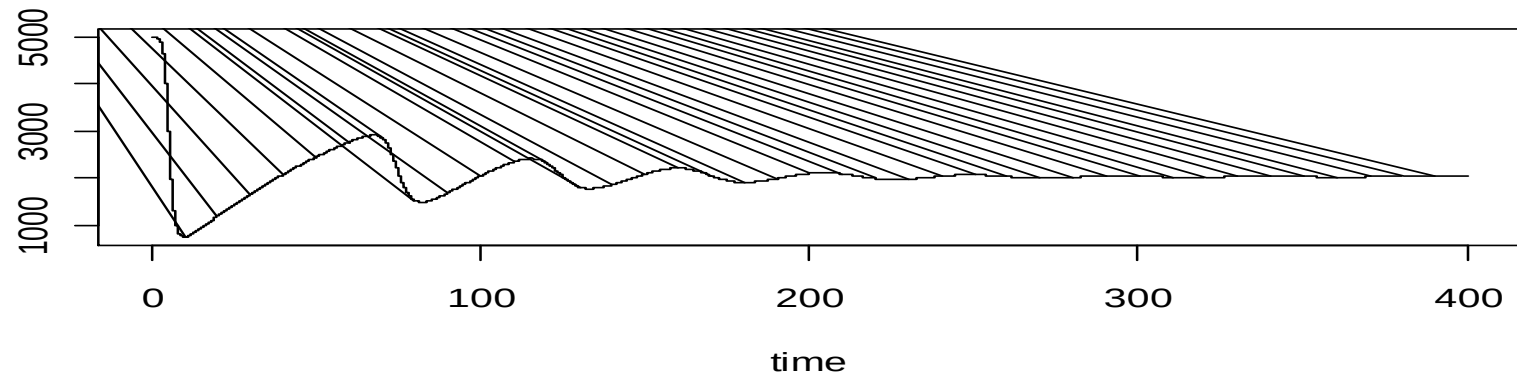
- Time for integration:

```
times<-seq(0,400,by=0.01)  
print(times[1:10])  
[1] 0.00 0.01 0.02 0.03 0.04 0.05 0.06 0.07 0.08
```

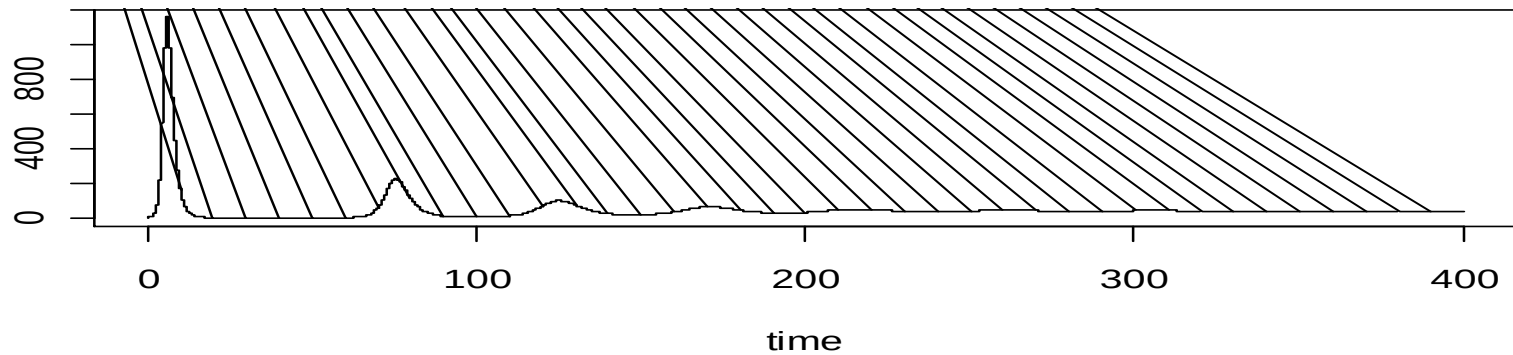
# Solution for the model

SIR model

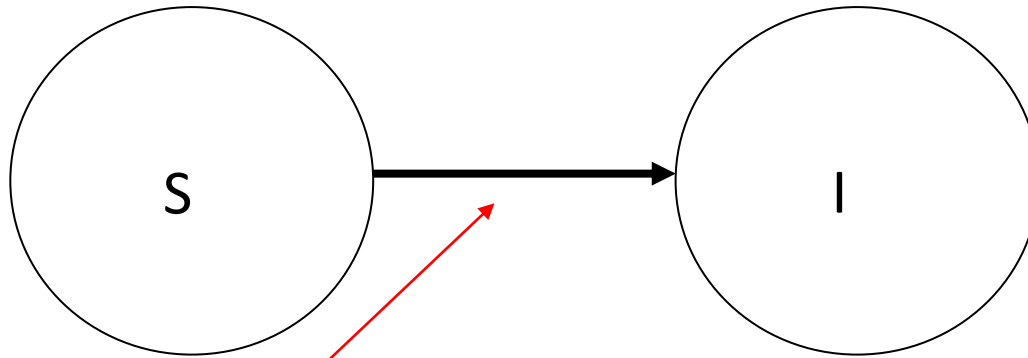
**S**



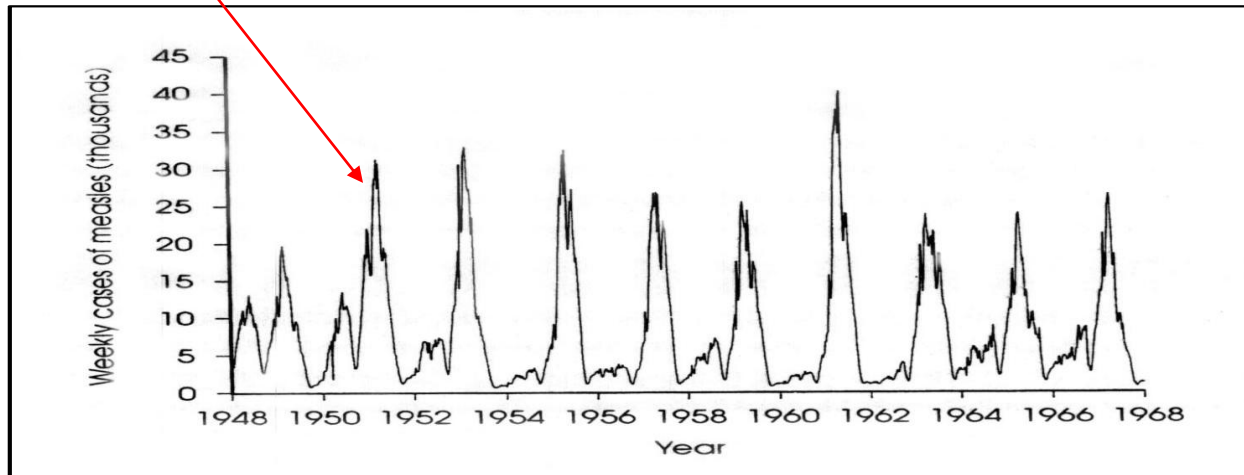
**I**



# Incidence data : Measles in UK

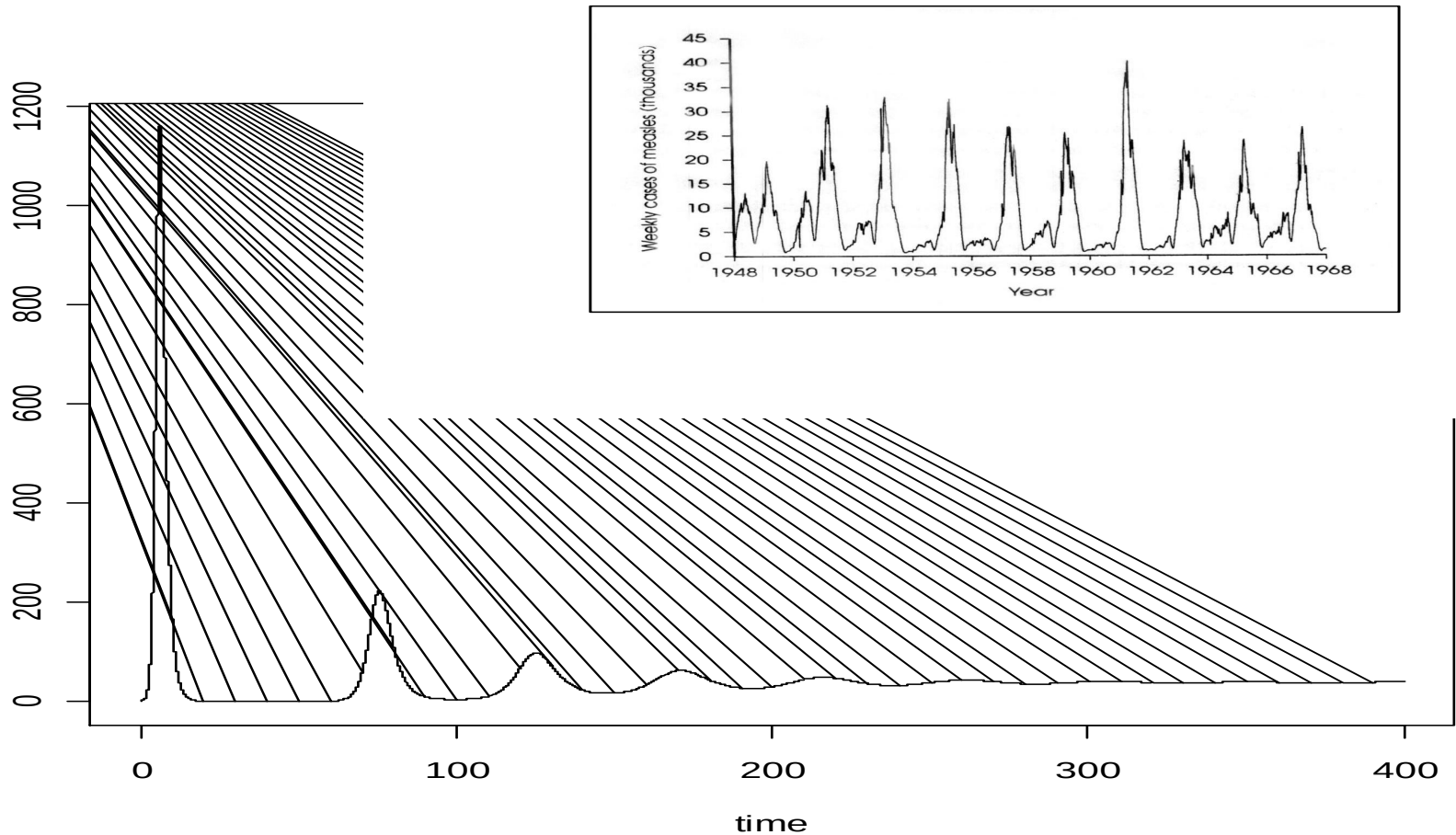


The **weekly** number of individuals who move from the susceptible to the infected class



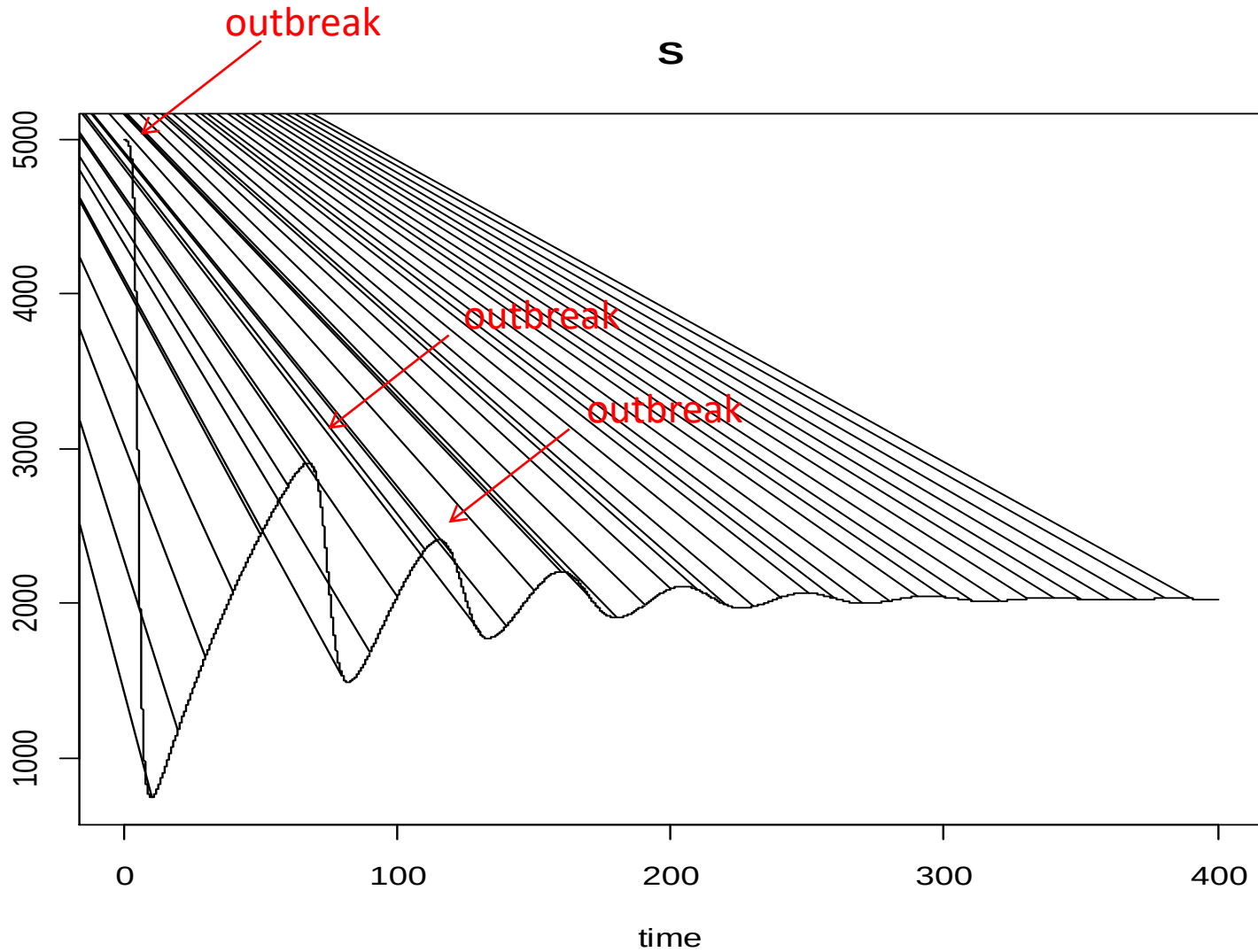
*Time unit=week*

# Observed outbreak and predicted outbreak

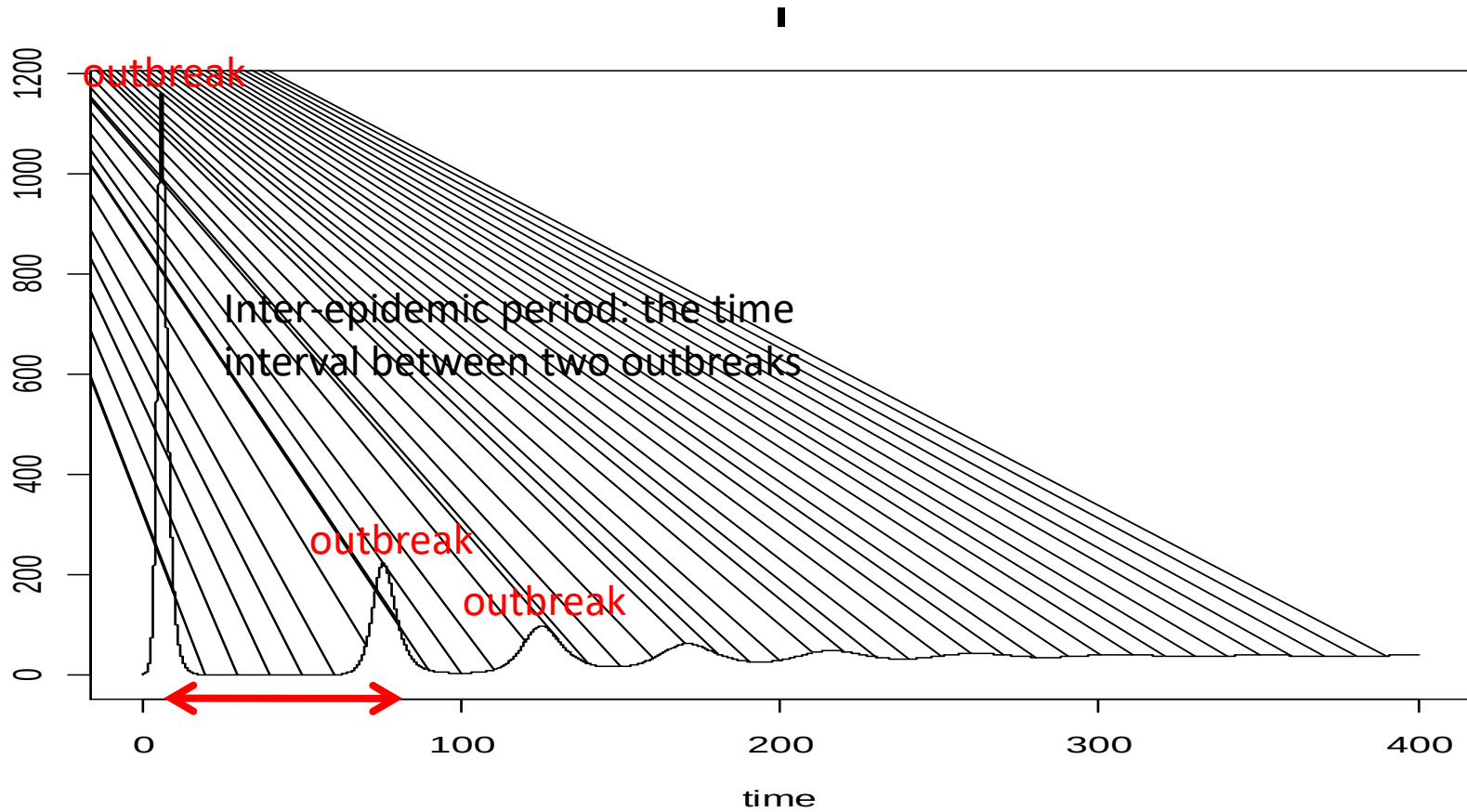


15

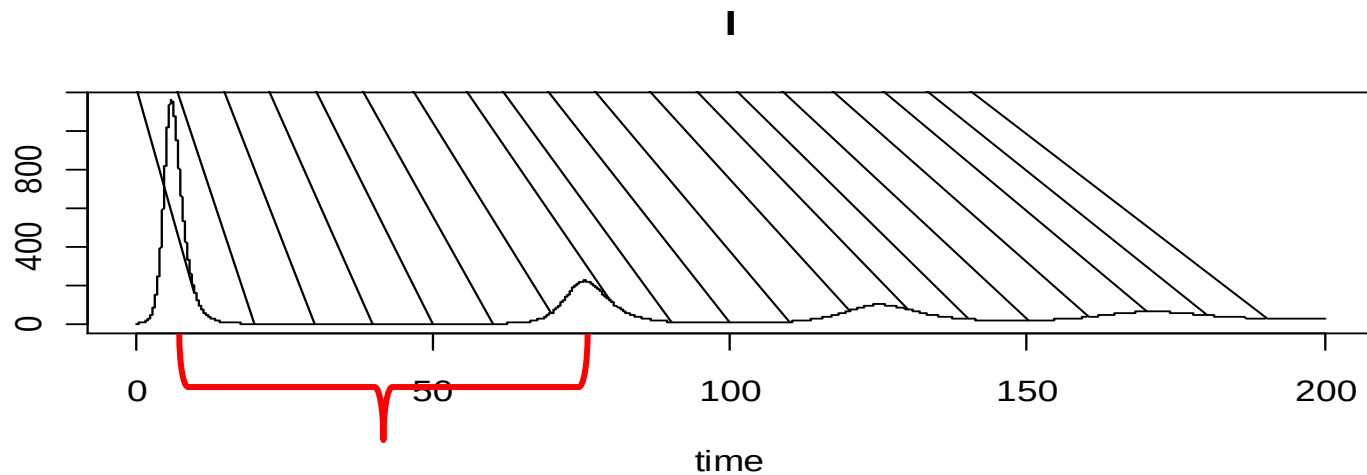
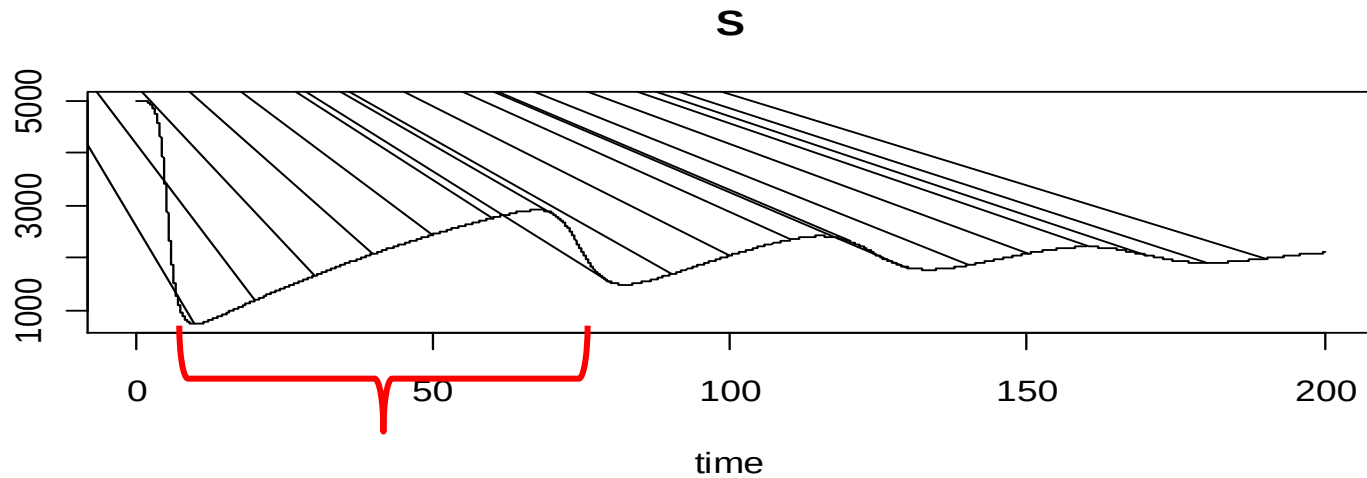
# The susceptible class



# The infected class

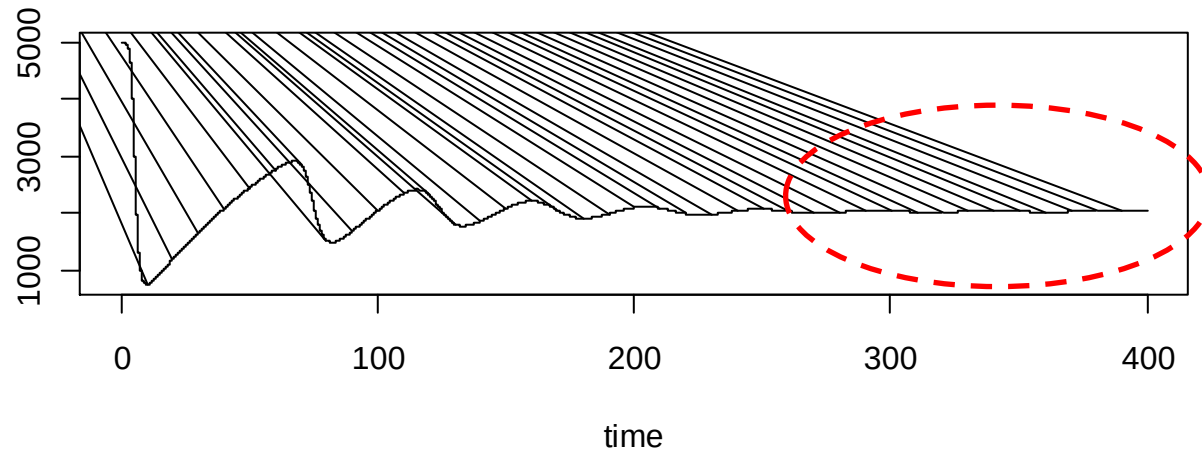


# The Inter epidemic period

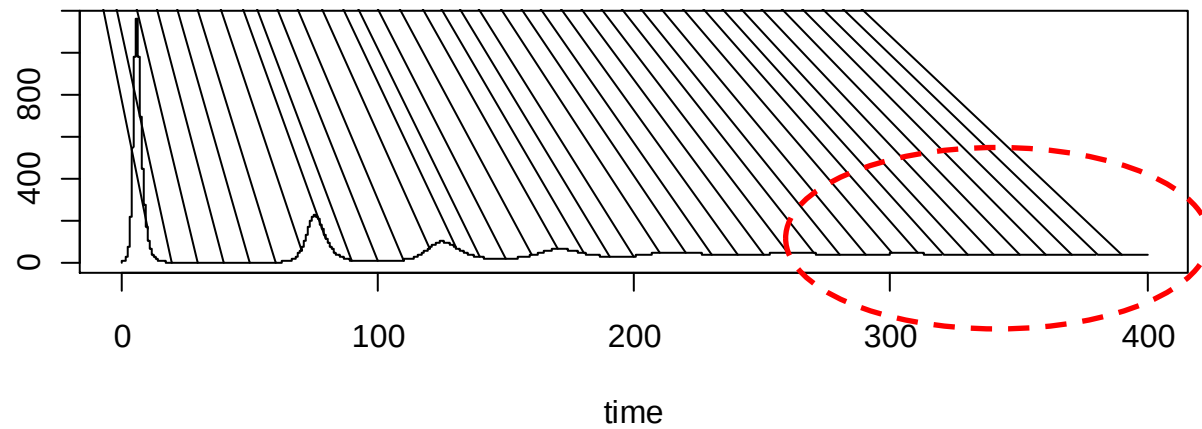


# Equilibrium

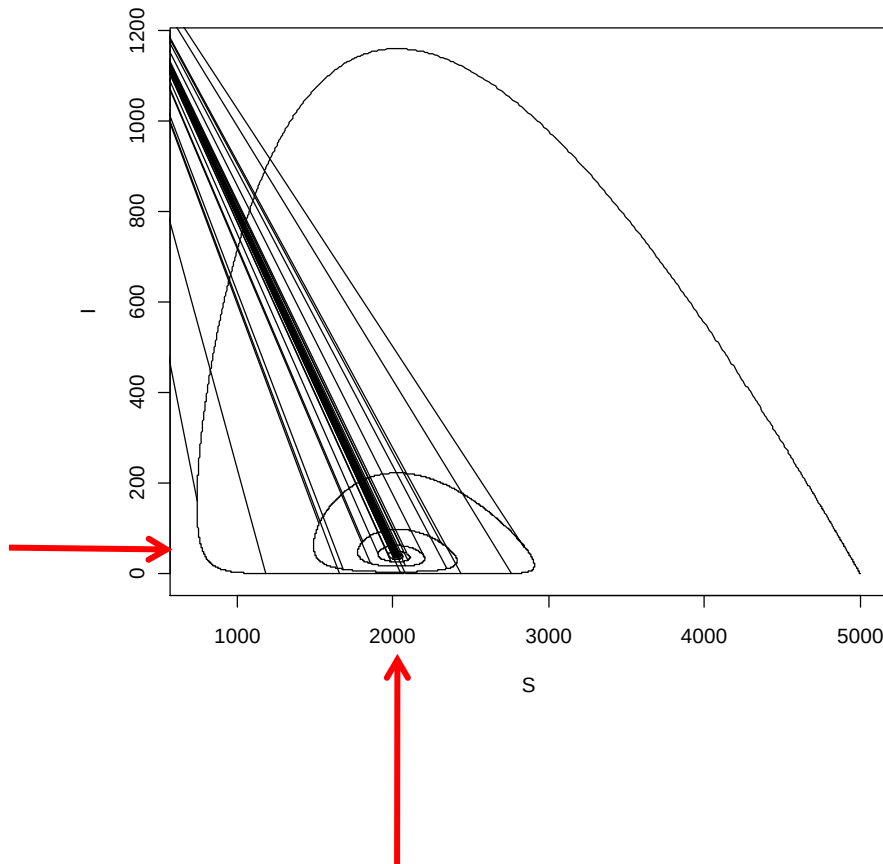
S



I



# Equilibrium



At the long run, the infection reach the endemic equilibrium state in which at each time unit there are the same number of susceptible and infected individuals in the population.

# Running the model in R

- To produce the model's output with **beta=0.001**:
  - Open R/R studio.
  - Open the program:
    - [ModelingIDinR1\\_V1\\_Stat&Dynam\\_10\\_2026.R](#).

# Slides 52-65: the HTML file (example 1)

C:/Ziv\_Temp\_2023/Tarylee\_STEAM-3D/Workshop1\_12\_10\_26/ShortCourse/Practicalessions\_New/er\_prog1\_STEAM3D\_2026-V1.html

er\_prog1\_STEAM3D\_2026-V1.html Open in Browser Find Publish

- 1. The SIR model
- 2. The SIR model in a closed population
- 3. The SIR model in open population**
  - Dynamic aspects of the SIR model
  - Equilibrium values
  - Example 1 ( $\beta=0.001$ )
  - Example 2: changing the transmission rate ( $\beta = 0.0005$ )
  - Example 3: changing population size
- 4. Threshold and Equilibrium Values

## 3. The SIR model in open population

### Dynamic aspects of the SIR model

We consider an open population model in which the birth rate is equal to the death rate,  $\mu(a, t)$ . We assume life long immunity and that there is no access mortality related to disease. The model formulated below follows individuals over time. This implies that the time unit of interest is the calendar time  $t$ .

$$\begin{aligned}\frac{\partial S(t)}{\partial t} &= N\mu(t) - (\lambda(t) + \mu(t))S(t), \\ \frac{\partial I(t)}{\partial t} &= \lambda(t)S(t) - (\nu + \mu(t))I(t), \\ \frac{\partial R(t)}{\partial t} &= \nu I(t) - \mu(a)R(t),\end{aligned}$$

The above model yields a constant population because of equal birth and death rates:

$$\frac{\partial S(t)}{\partial t} + \frac{\partial I(t)}{\partial t} + \frac{\partial R(t)}{\partial t} = \mu(N - S(t) - I(t) - R(t)) = 0.$$

### Equilibrium values

The equilibrium values,  $S_\infty, I_\infty, R_\infty$  are the number of individuals in each compartment when  $t \rightarrow \infty$  and given by

- $S_\infty = \frac{\nu + \mu}{\beta} \times N$ .
- $I_\infty = \frac{\mu}{\beta} (R_0 - 1)$ .
- $R_\infty = N - S_\infty - I_\infty$ .

See Section 4 below for a numerical example.

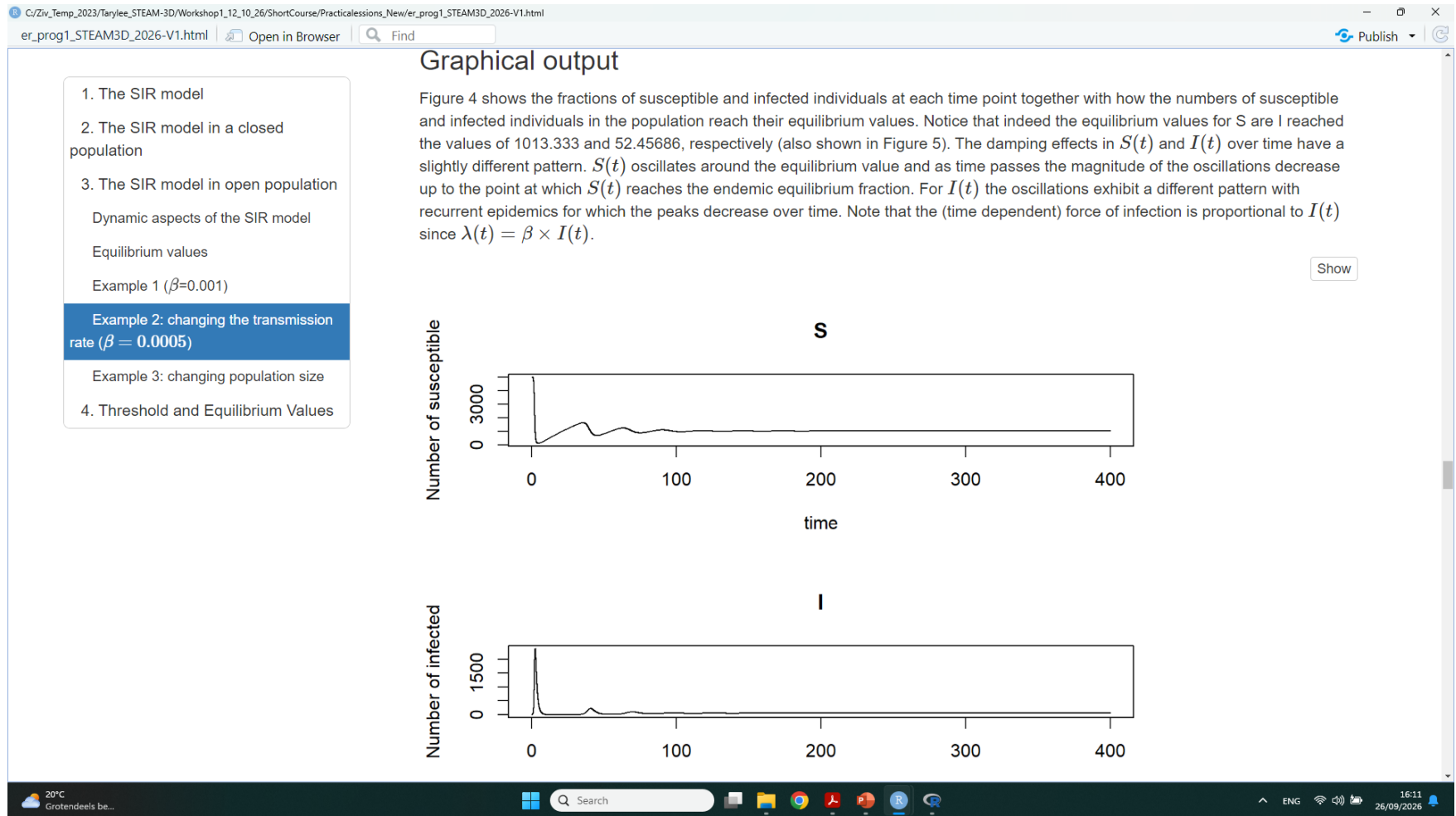
**Example 1 ( $\beta=0.001$ )**

20°C Grotendeels be...

Search

ENG 16:10 26/09/2026

# The HTML file: model's solutions



# Changing the parameters setting

- We consider two examples in which we change the parameters setting:

1. How the transmission changes when we change the transmission rate ?

$$\beta = 0.001 \rightarrow \beta = 0.0005$$

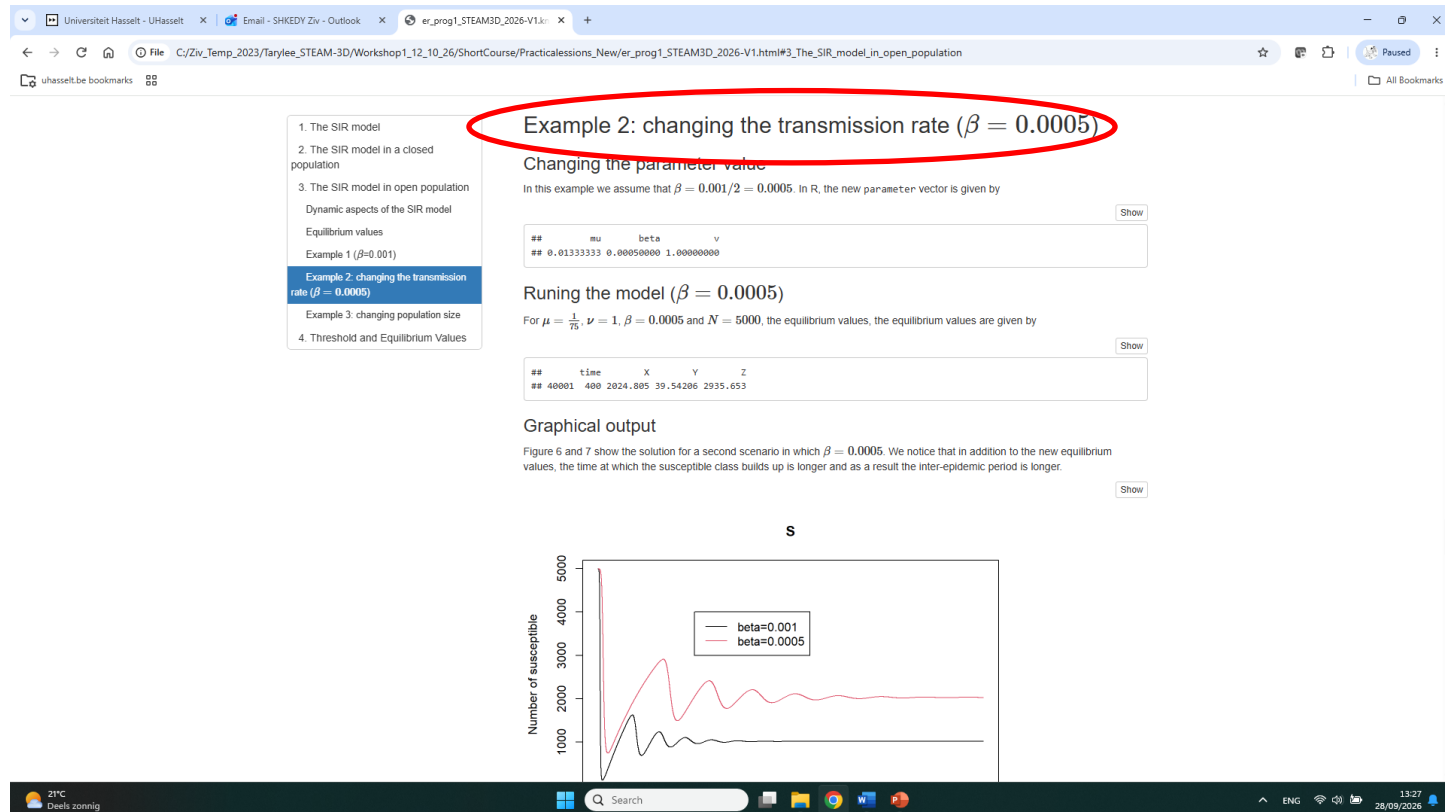
2. How the transmission changes when we change the population size ?

$$N = 5000 \rightarrow N = 2500$$

# Changing the parameter setting (example 2)

- Changing the transmission rate:

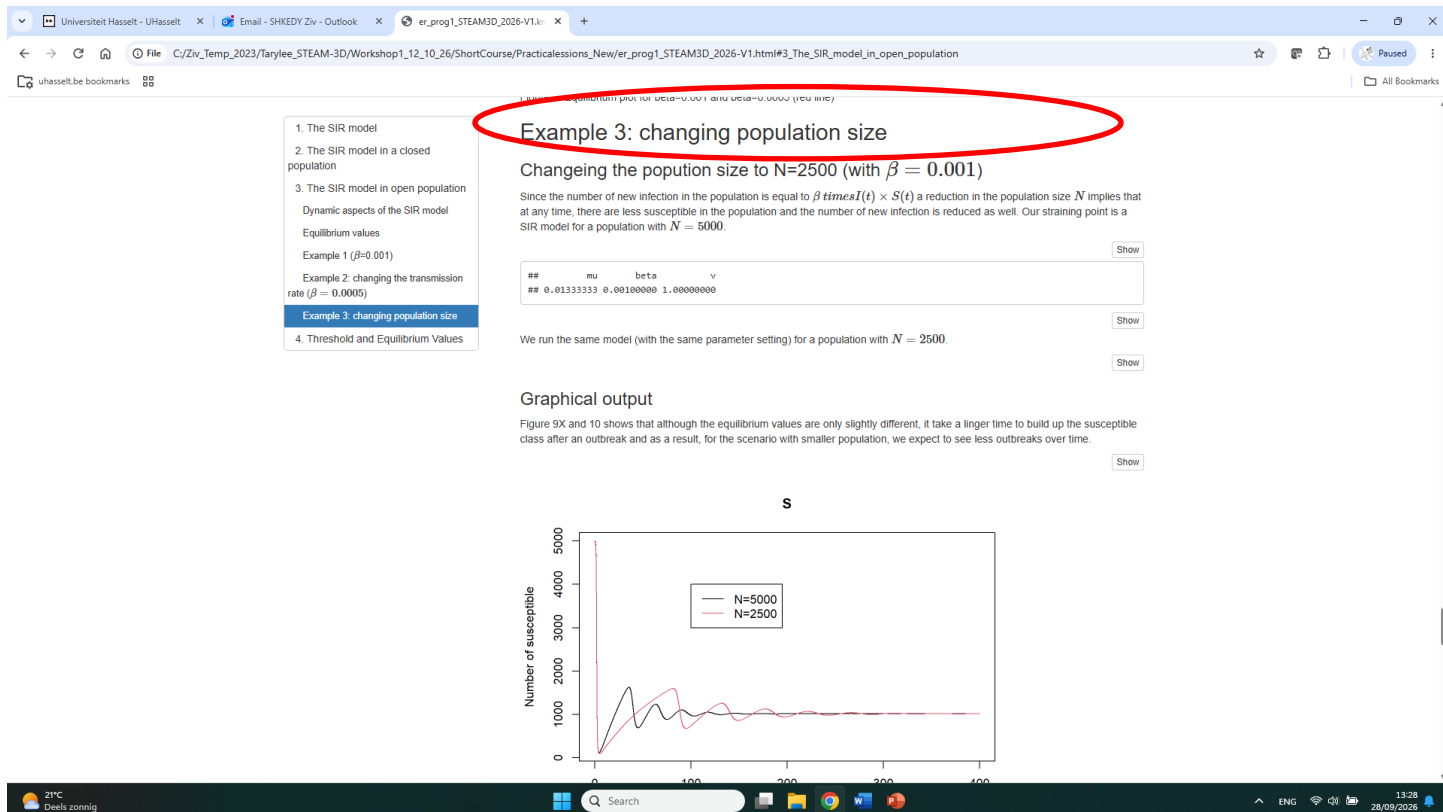
$$\beta = 0.001 \rightarrow \beta = 0.0005$$



# Changing the parameter setting (example 3)

- Changing the population size:

$$N = 5000 \rightarrow N = 2500$$



# Running the model in R

- Make the two examples by yourself.
- Look first at the code and output in the HTML.
- To produce the model's output:
  - Open R/R studio.
  - Open the program:
    - [ModelingIDinR1\\_V1\\_Stat&Dynam\\_10\\_2026.R.](#)

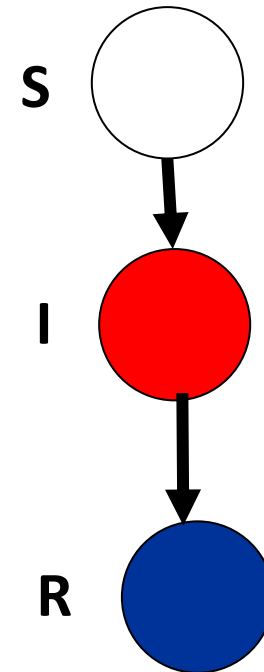
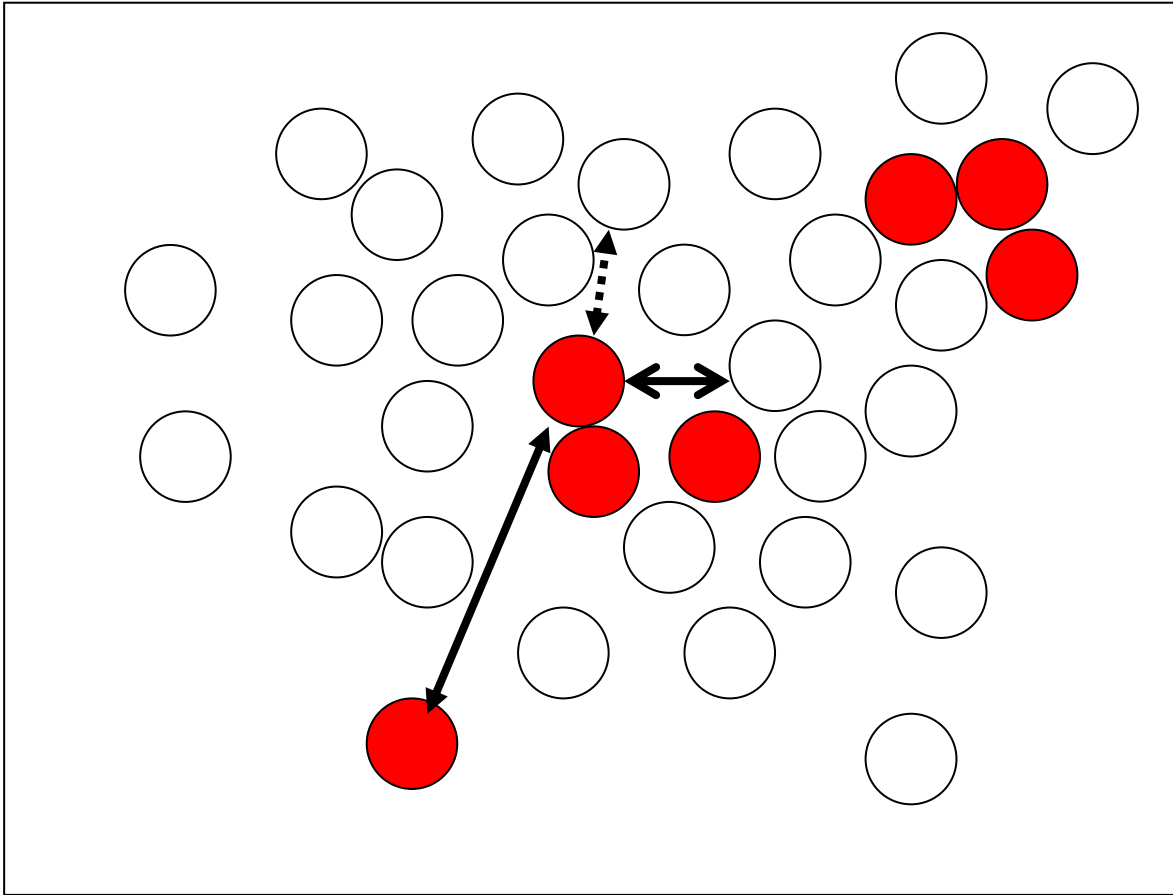
## Part 4:

# Vaccination in SIR model

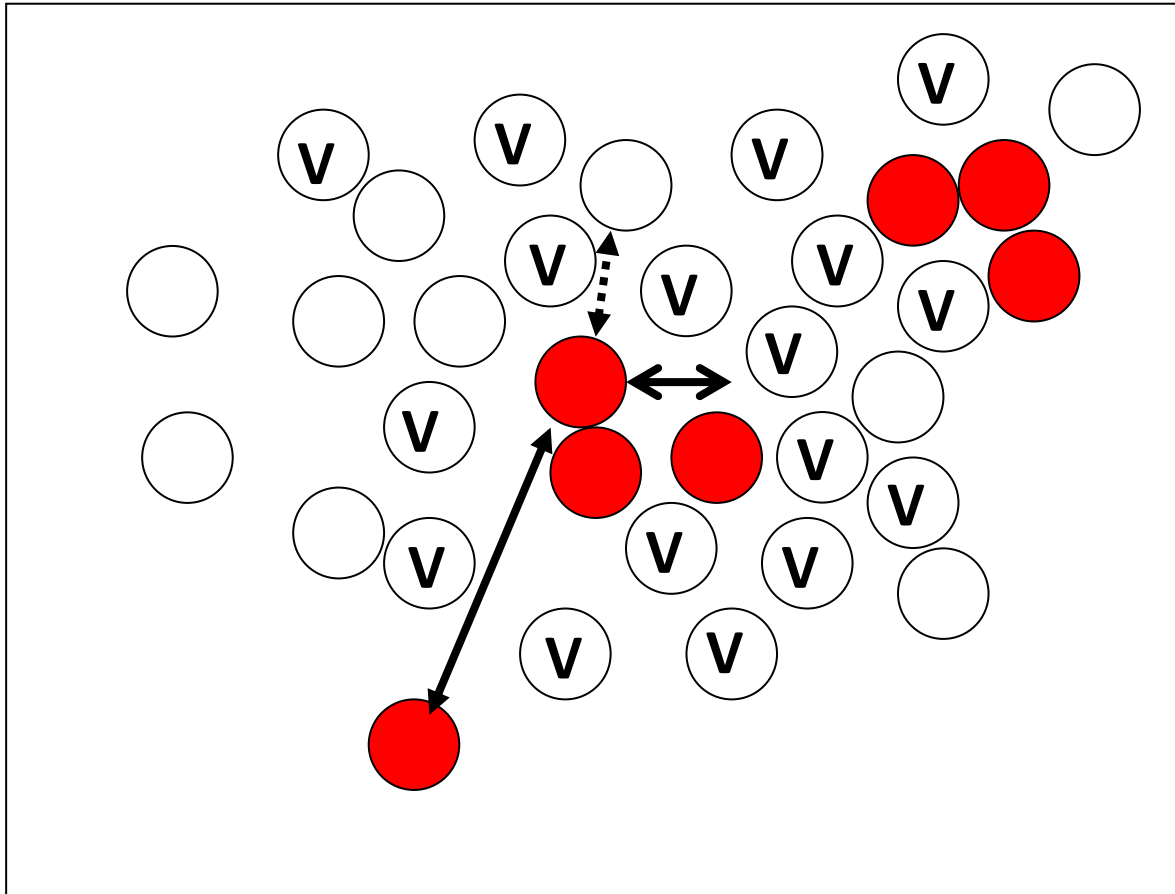
# R programs for the practical session

- HTML file for slides 81-99:
  - [eR\\_prog2\\_STEAM3D\\_2026\\_V1.html](#)
- Two R programs:
  - R program:
    - [ModelingIDinR4\\_V1\\_Vaccination\\_10\\_2026.R](#).
    - R requirements: basic R editor.
  - R Markdown:
    - [eR\\_prog2\\_STEAM3D\\_2026\\_V1.Rmd](#).
    - R requirements: R Studio.
    - This program produces the HTML file

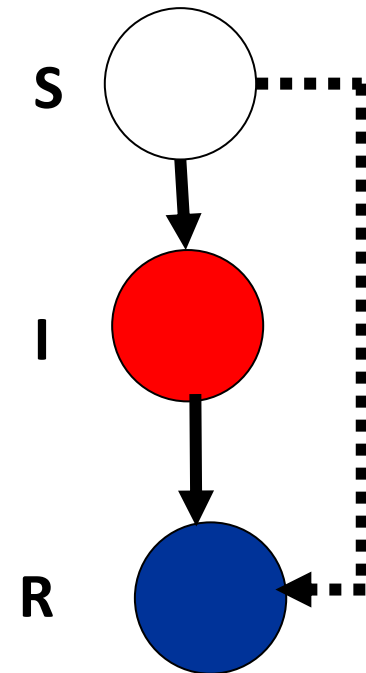
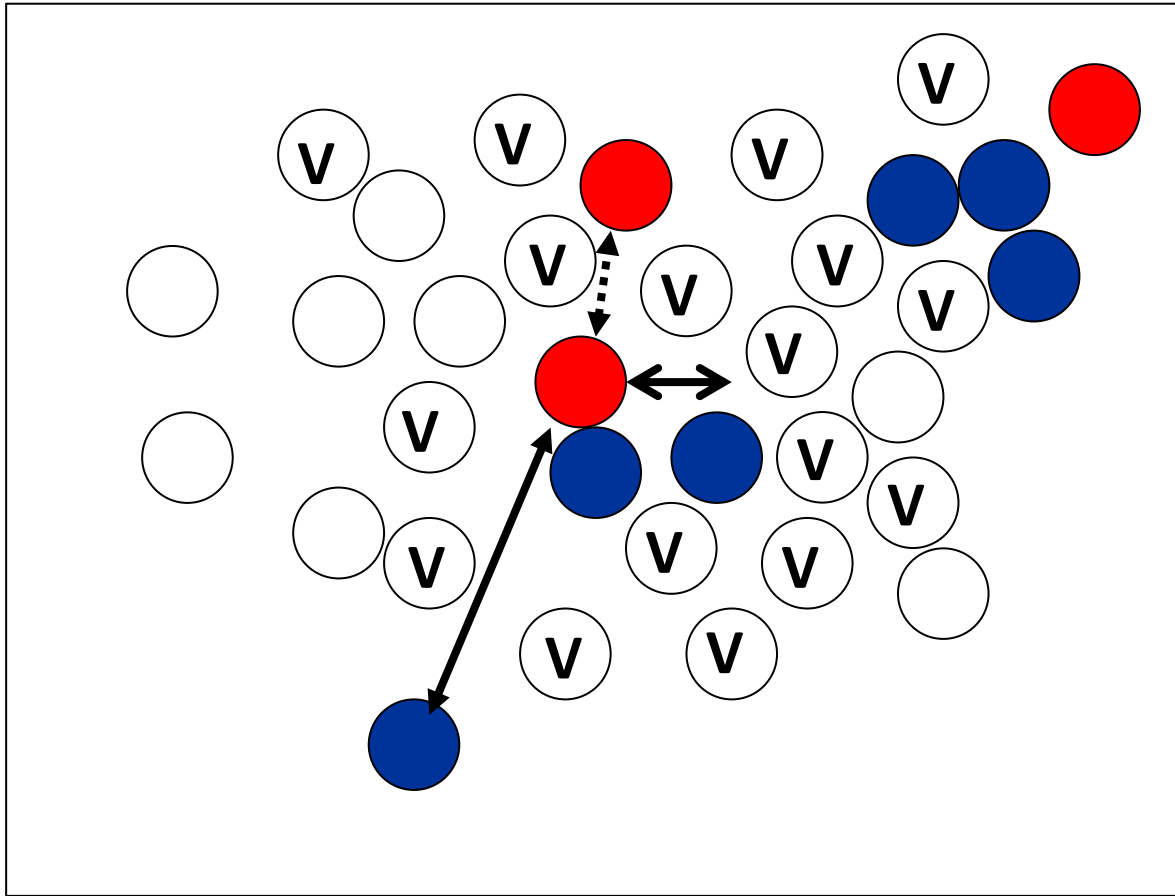
# Vaccination



# Vaccination

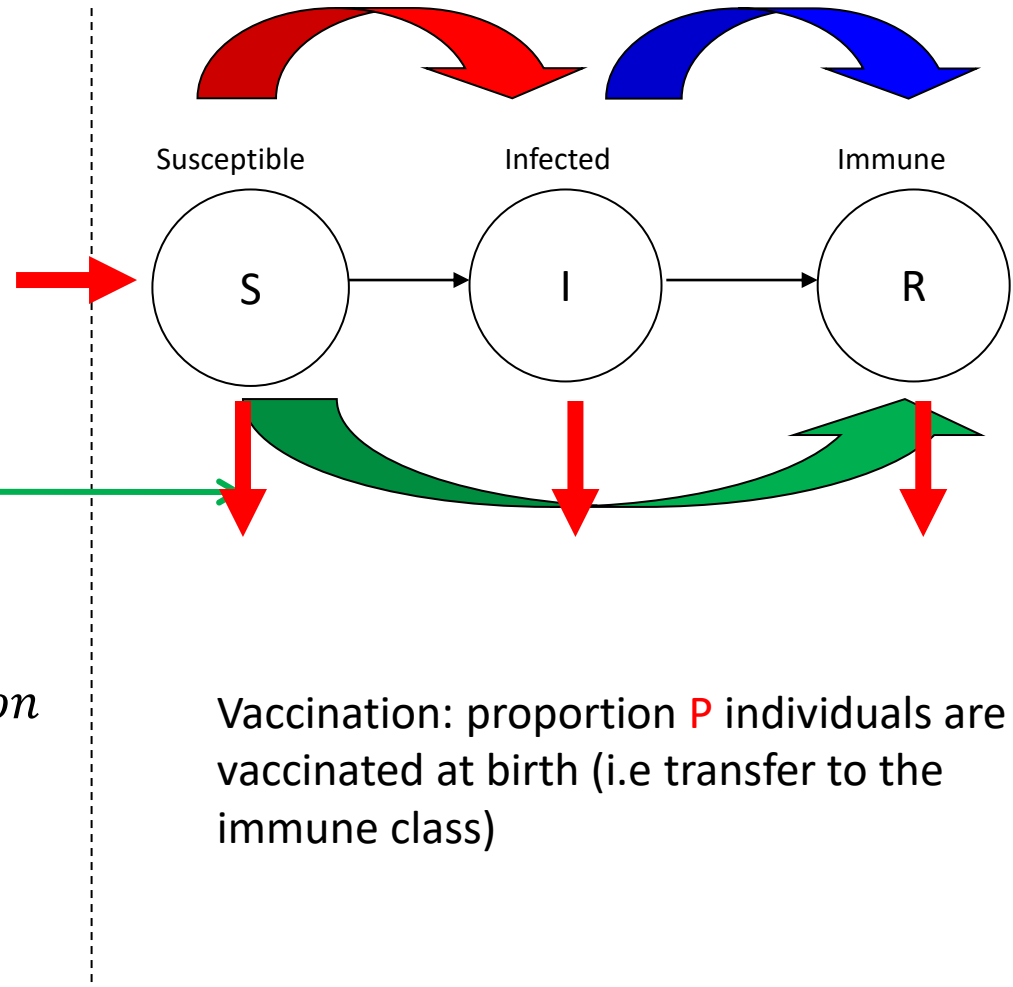


# Vaccination



# Transmission model with vaccination

$$\begin{aligned}\frac{dS(t)}{dt} &= N\mu \times (1 - P) - \lambda S - \mu S \\ \frac{dI(t)}{dt} &= \lambda S - (\sigma + \mu)I \\ \frac{dR(t)}{dt} &= \sigma I - \mu R + N\mu \times P\end{aligned}$$



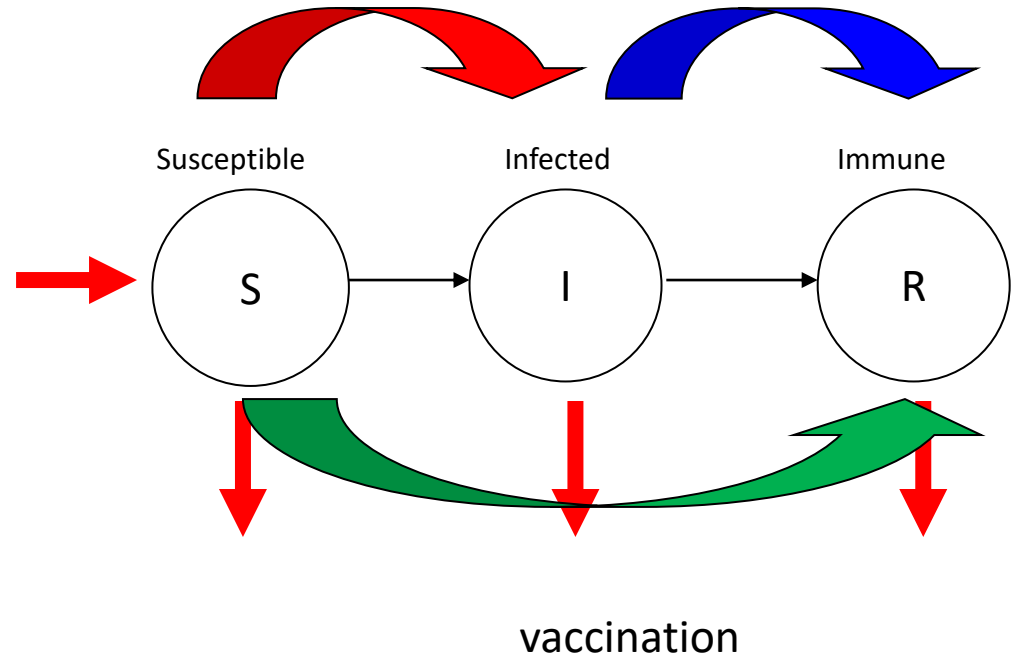
$\lambda(t) = \beta \times I(t)$ : force of infection

Vaccination: proportion **P** individuals are vaccinated at birth (i.e transfer to the immune class)

# SIR transmission Model with vaccination

Model parameters:

1. Life expectancy: 75 years.
2. Transmission rate 0.0005.
3. Recovery rate 1 time unit.
4. Vaccination **45%** at birth.

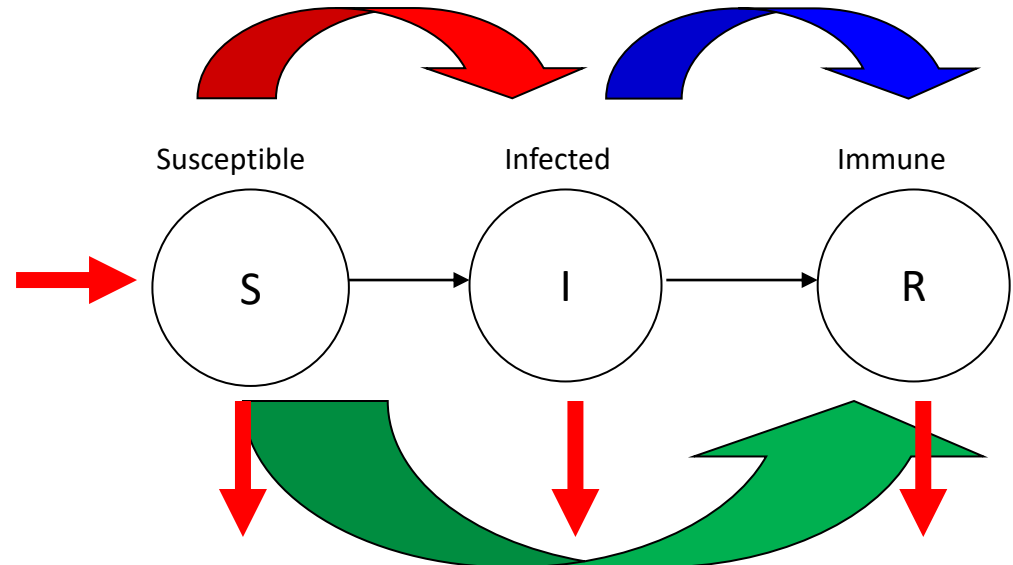


```
> parameters <- c(mu=1/75,beta=0.001/2, v=1, P=0.45)
> parameters
      mu      beta      v      P
0.01333333 0.00050000 1.00000000 0.45000000
```

# SIR transmission model with vaccination

- The model in R:

```
SIR<-function(t,state,parameters)
{
  with(as.list(c(state, parameters)),
  {
    dX <- 5000*mu*(1-P)-beta*Y*X - mu*X
    dY <- beta*Y*X - v*Y - mu*Y
    dZ <- v*Y -mu*Z+5000*mu*P
    list(c(dX, dY, dZ))
  })
}
```



$\beta$  = transmission rate.  
 $N$  = population size.  
 $\mu$  = birth rate = death rate.  
 $v$  = recovery rate.  
 $p$  = vaccinate rate.

$$\left\{ \begin{array}{l} \frac{dS(t)}{dt} = N\mu \times (1 - P) - \beta IS - \mu S \\ \frac{dI(t)}{dt} = \beta IS - (v + \mu)I \\ \frac{dR(t)}{dt} = vI - \mu R + N\mu \times P \end{array} \right.$$

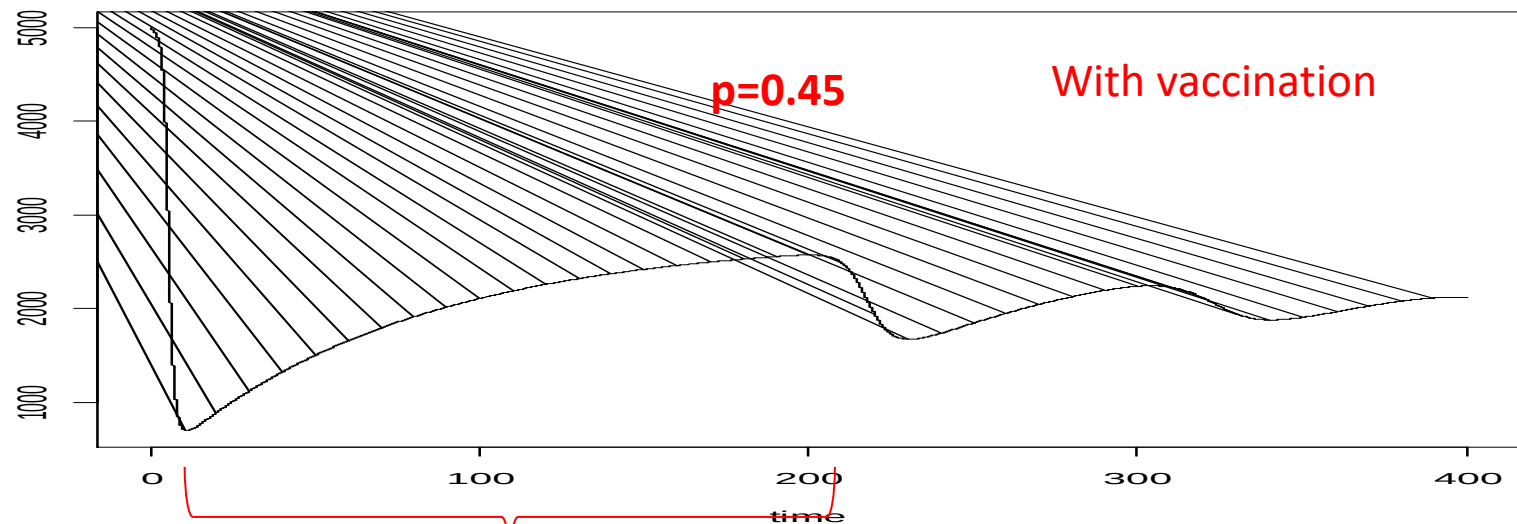
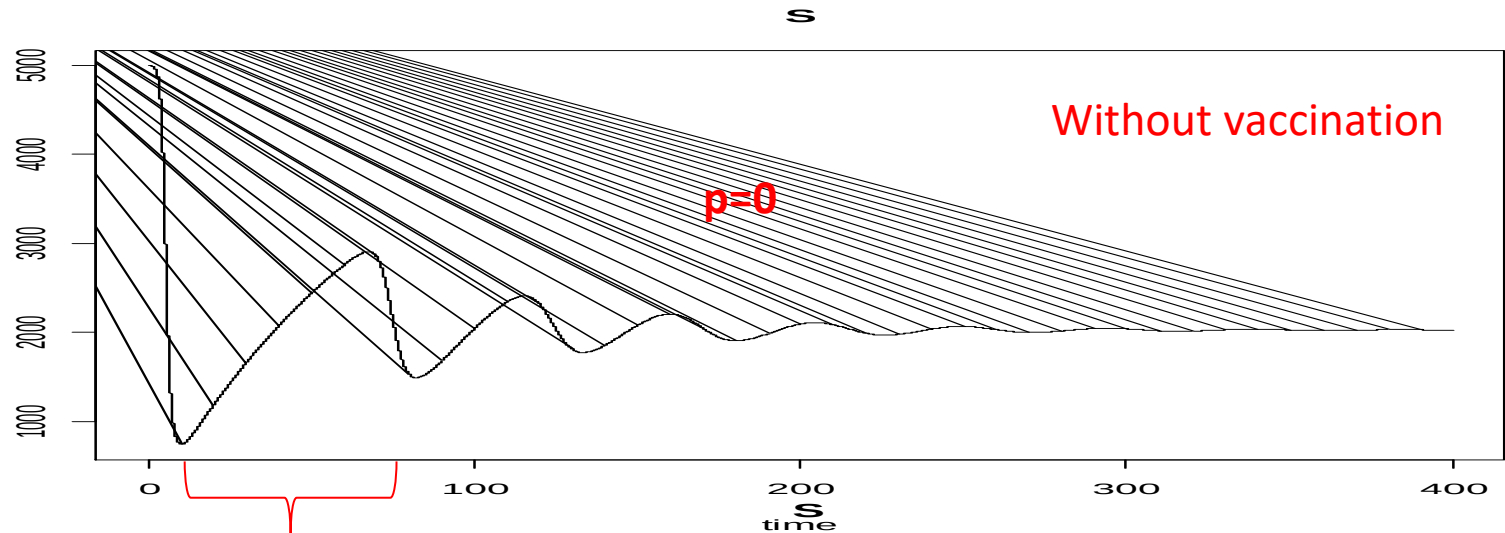
Part 4.1:  
Example 1: SIR model with  $p=0.45$

# SIR model with vaccination at birth

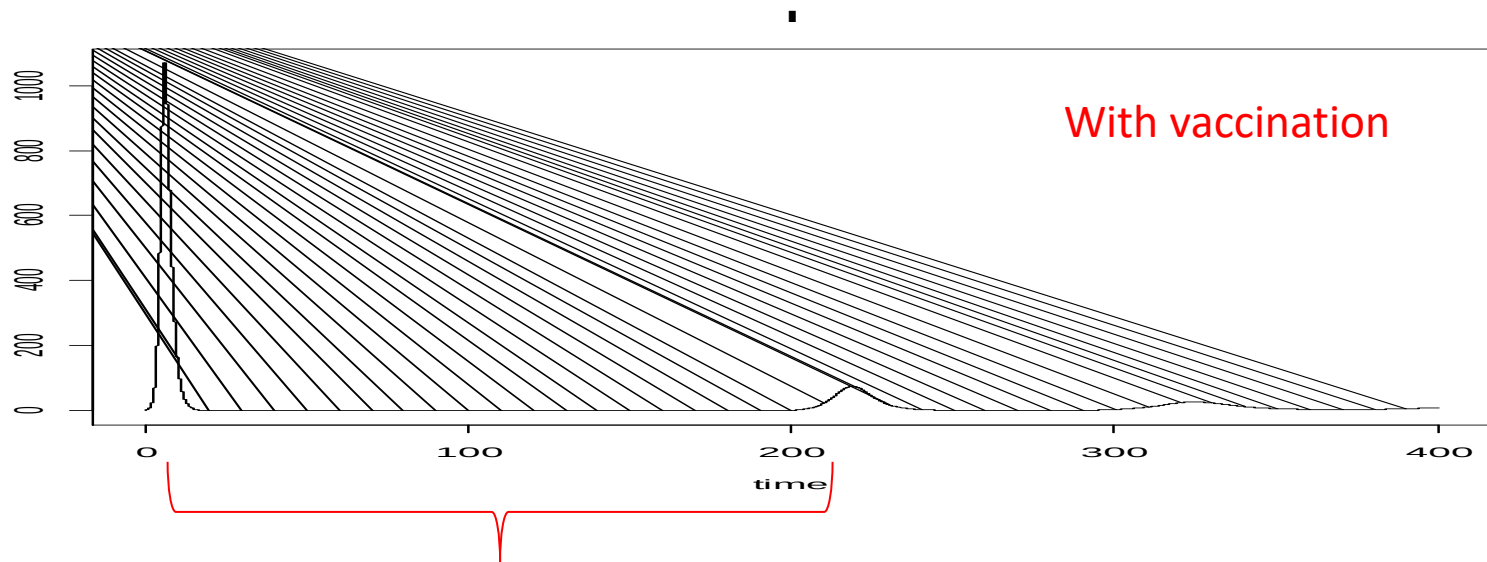
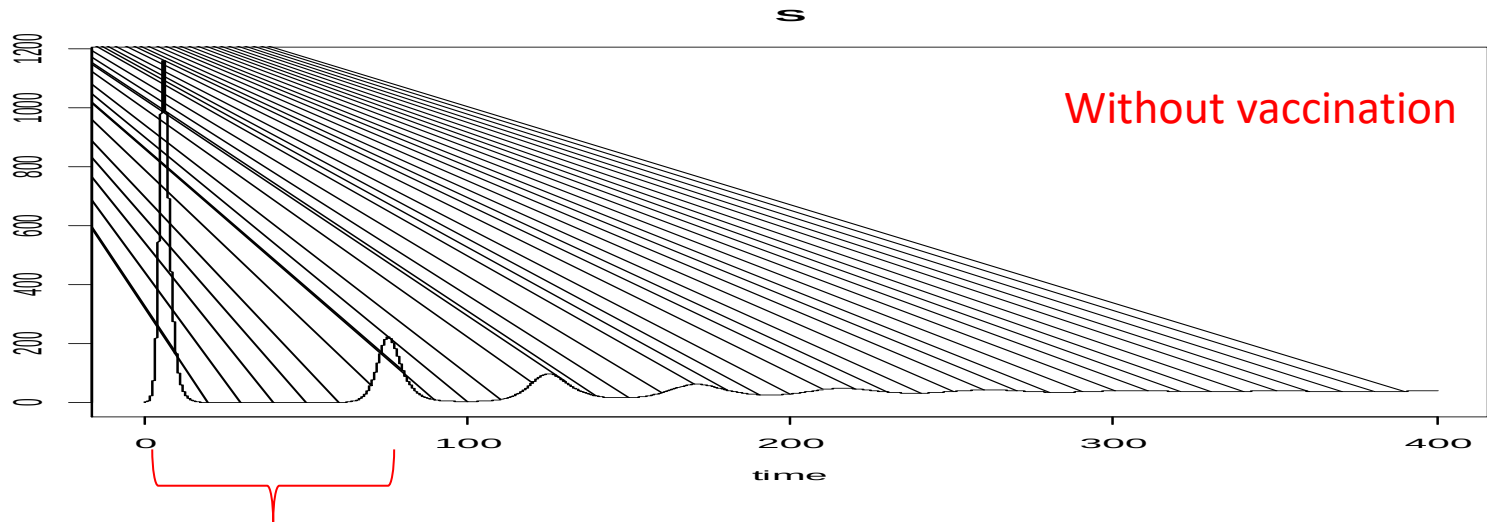
**45%** are vaccinated at birth.

```
> parameters <- c(mu=1/75,beta=0.001/2, v=1, P=0.45)  
> parameters  
           mu           beta           v           P  
0.01333333 0.00050000 1.00000000 0.45000000
```

# Susceptible: inter epidemic period

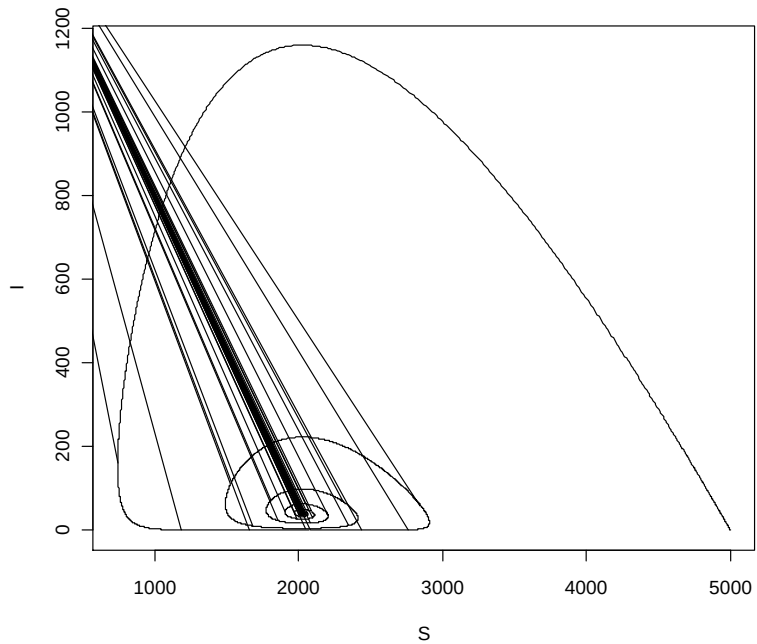


# Infected class: inter epidemic period

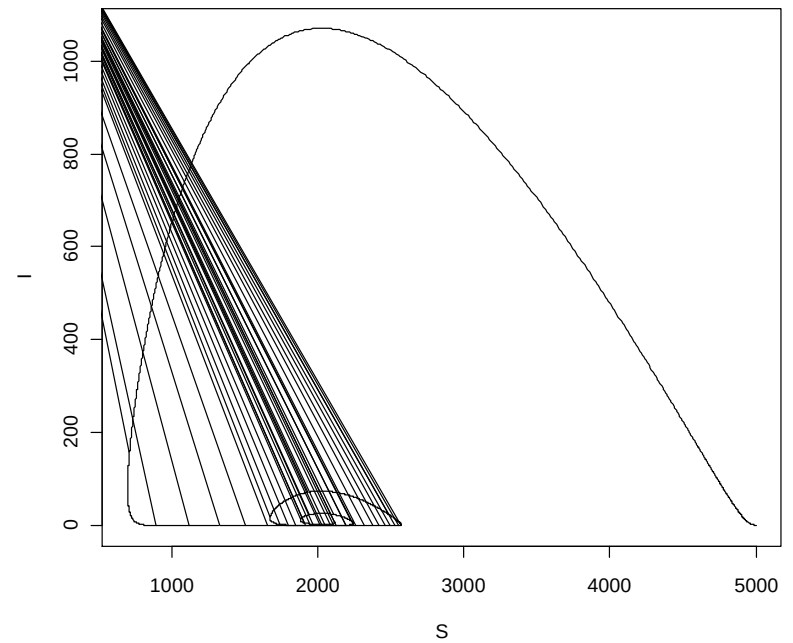


# Equilibrium

Without vaccination



With vaccination



## Part 4.2:

Example 1: SIR model with  $p=0$  and  $p=0.4$

# Parameter setting

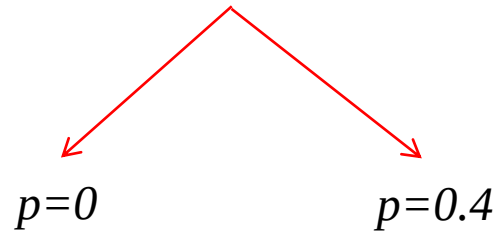
Two scenarios:

$$\beta = 0.0005.$$

$$N=5000.$$

$$\mu = \frac{1}{75}.$$

$$v = 1.$$



# Parameter setting in R

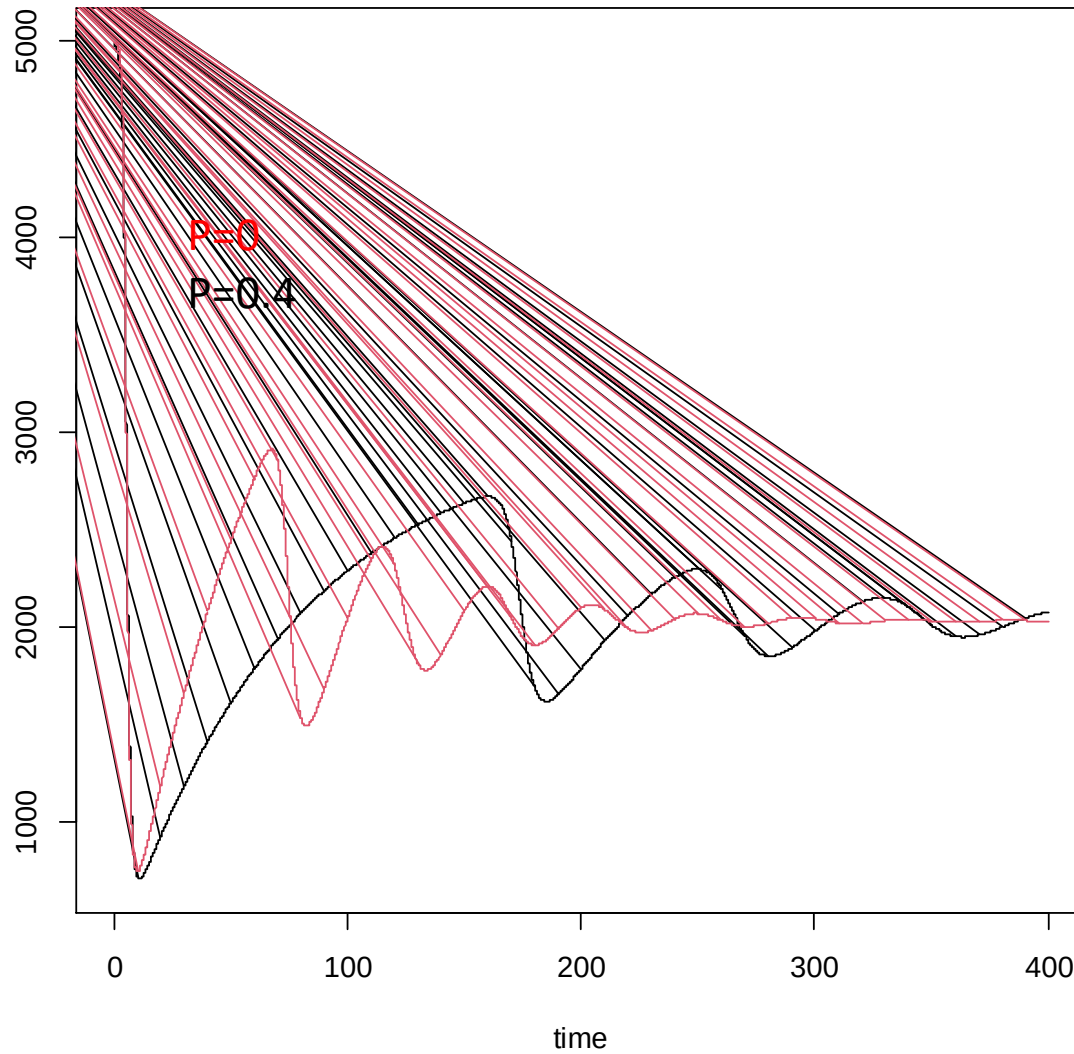
```
> stateparameters <- c(mu=1/75,beta=0.001/2, v=1, P=0.40)
> parameters
      mu      beta      v      P
0.01333333 0.00050000 1.00000000 0.40000000
> state <- c(X=4999,Y=1,Z=0)
> state
   X    Y    Z
4999  1    0
>
```



$S(0) = 4999, I(0) = 1, R(0) = 0$

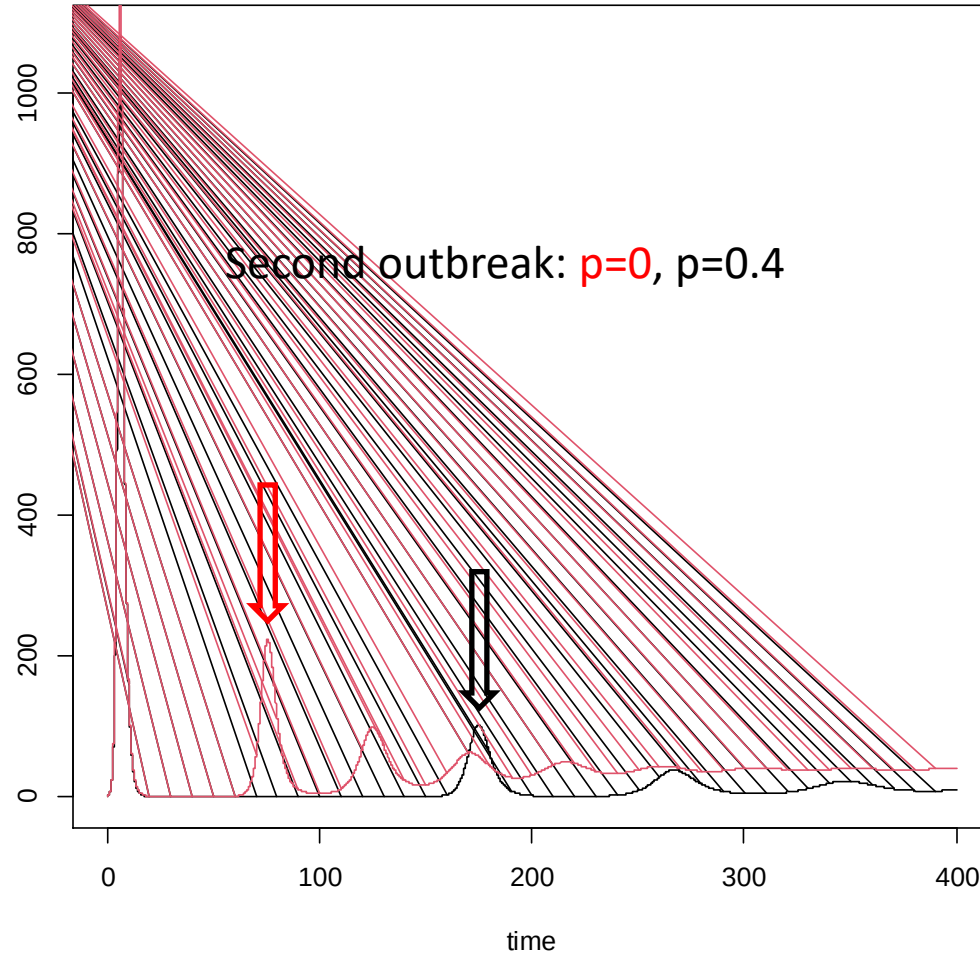
# Susceptible

s

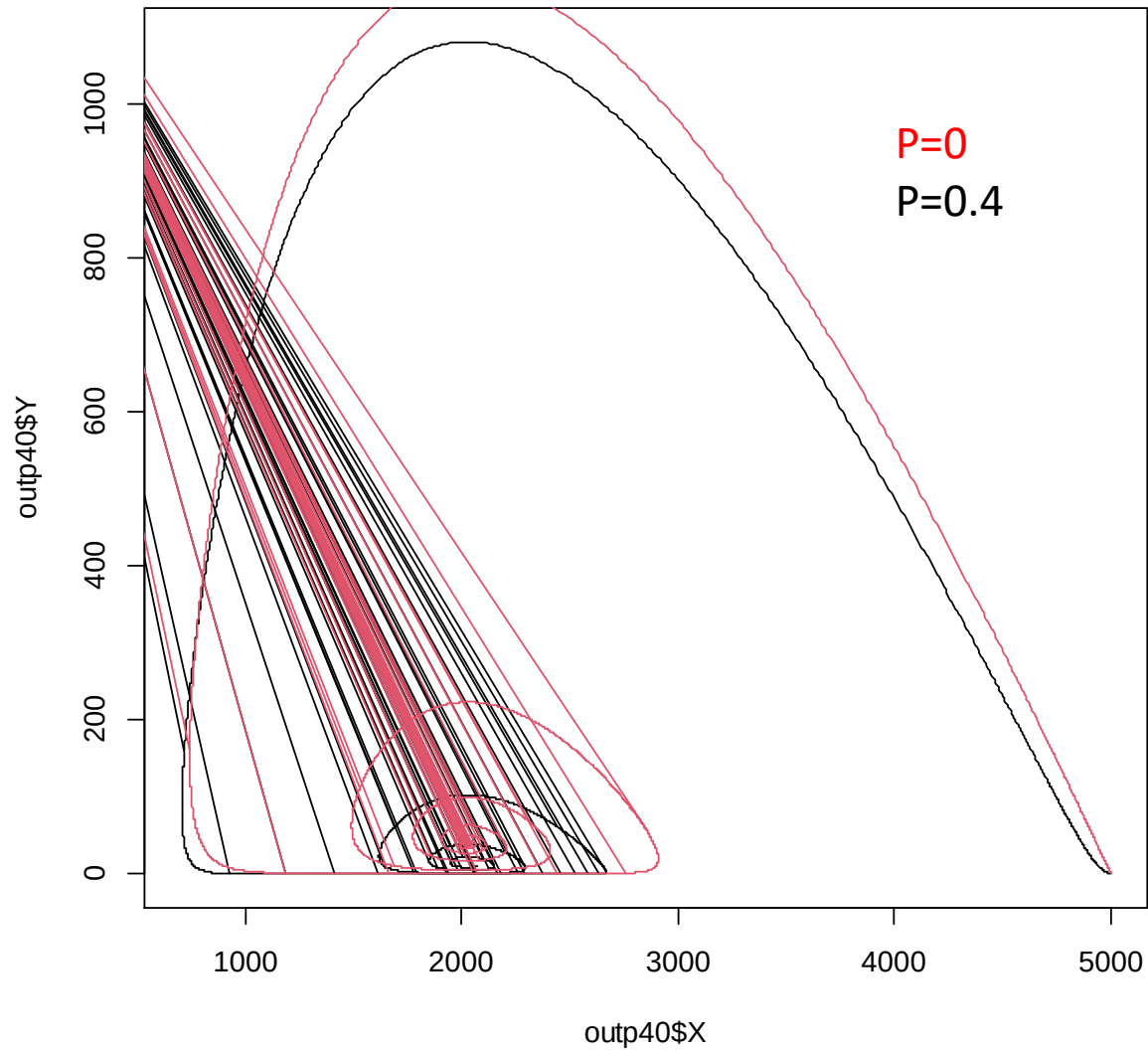


# Infected

I



# Equilibrium values



# Slides 92-97: the HTML file

er\_prog2\_STEAM3D\_2026-V1.ko x +

C:/Ziv\_Temp\_2023/Tarylee\_STEAM-3D/Workshop1\_12\_10\_26/ShortCourse/Practicalessions\_New/er\_prog2\_STEAM3D\_2026-V1.html

uhasselt.be bookmarks

Code

24-09-2026 >eR-BioStat and STAEM-3D

## Modeling infectious diseases using R - vaccination in the SIR model

Ziv Shkedy and Rashider Aloni based on Chapter 2-4 in the book *Modeling infectious diseases parameters based on serological and social contact data* (<https://www.simid.be/modeling-infectious-disease-parameters-based-on-serological-and-social-contact-data/>)

Show

### 1. The SIR model with vaccination in open population

Incorporating a vaccination program in the SIR model can be done by adding a direct path from the susceptible compartment to the immune compartment. We illustrate this using a SIR model that is age-independent and we look at its dynamic behavior over time. Let us assume that a proportion  $p$  of each cohort of newborns is vaccinated at birth, the SIR model with demography can then be written as

$$\begin{cases} \frac{dS}{dt} = N\mu(1-p) - (\lambda(t) + \mu)S, \\ \frac{dI}{dt} = (\lambda(t) + \mu)S - (\nu + \alpha + \mu)I, \\ \frac{dR}{dt} = N\mu p + \nu I - \mu R. \end{cases}$$

The SIR model formulated above implies that, for each cohort, a proportion  $p$  enter the population directly in the immune class and therefore do not take a part in the transmission process.

### 2. The SIR model with vaccination in R: post vaccination equilibrium

Recall that the basic reproductive number,  $R_0$ , is defined as

$$R_0 = \frac{N \times \beta}{\nu + \mu}.$$

If  $R_0$  is above the threshold of one, the disease will maintain itself while for  $R_0 < 1$  the disease will die out.

### The Critical Proportion of Vaccination

20°C Grotendeels be...

Search

ENG 16:51 26/09/2026

# Slides 92-97: the HTML file - example 2

er\_prog2\_STEAM3D\_2026-V1.k... X +

C:/Ziv\_Temp\_2023/Tarylee\_STEAM-3D/Workshop1\_12\_10\_26/ShortCourse/Practicalessions\_New/er\_prog2\_STEAM3D\_2026-V1.html

uhaselt.be bookmarks

- 1. The SIR model with vaccination in open population
- 2. The SIR model with vaccination in R: post vaccination equilibrium
- 3. Example 1: a model without vaccination ( $p = 0$ )
- 4. Example 2: model with 40% vaccination at birth**
- 5. Example 3: The critical proportion of vaccination

## 4. Example 2: model with 40% vaccination at birth

### Definition of the model parameters

In this example we assume that 40% of the population are vaccinated at birth. The state variables remain the same as before

Show

```
##      X      Y      Z
## 4999      1      0
```

but we need to specify the proportion of vaccination. For a population size of 5000,  $\beta = 0.0005$ ,  $\nu = 1$ , life expectancy of 75 years and vaccination rate of 40% at birth,  $p = 0.4$  we have

Show

```
##      mu      beta      v      P
## 0.01333333 0.00050000 1.00000000 0.40000000
```

### Model solution

Figure 3 shows the number of susceptible over time for two scenarios with  $p = 0$  and  $p = 0.4$  and reveals the post vaccination equilibrium patterns. The inter epidemic period is longer when the proportion of vaccinated individuals increases. This implies that individuals stay, on average, longer time in the susceptible class and as a result the average age in which individuals are infected (the average age at infection) increases. In the long run, after vaccination, the population will research a new endemic equilibrium. Notice that although the pre-vaccination force of infection  $\lambda(0)$  is equal under the two scenarios, in the new equilibrium the force of infection decreases as the proportion of vaccinated individuals increases. Since  $\lambda(t)$  decreases as  $p$  increases the rate in which individuals leave the susceptible class decreases as  $p$  increases and as a result, as shown in Figure 4 the inter epidemic period is longer as  $p$  increases. This implies that a vaccination program has two effects on the population:

- The direct effect is the transfer of a proportion  $p$  from the susceptible directly to the immune class.
- The indirect effect is related to the decline of the magnitude of the force of infection and its affect both vaccinated and unvaccinated individuals.

Show

```
##      time      X      Y      Z
## 1 0.00 4999.000 1.000000 0.000000
## 2 0.01 4998.708 1.014972 0.2767231
## 3 0.02 4998.416 1.030167 0.5535601
## 4 0.03 4998.124 1.045588 0.8305133
## 5 0.04 4997.831 1.061237 1.1075846
## 6 0.05 4997.538 1.077120 1.3847767
```

Show

20°C Grotendeels be...

Search

ENG 16:53 26/09/2026

# Running the model in R

- Make the two examples (with  $p=0$  &  $p=0.4$ ) by yourself.
- Look first at the code and output in the HTML file.
- To produce the model's output:
  - Open R/R studio.
  - Open the program:
    - [ModelingIDinR4\\_V1\\_Vaccination\\_10\\_2026.R](#).